

Computer Empathy

Understanding how computers think to enable business professionals to better leverage the power of these general purpose machines

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Last Revised – February 18, 2019 (DRAFT)

“He who loves practice without theory is like the sailor who boards a ship without a rudder and compass and never knows where he may cast.” *Leonardo da Vinci*

Executive summary:

- Changes that will be brought about by what is being called “the artificial intelligence revolution” among other terms will unquestionably be significant. It is not useful to overstate or understate the impact¹.
- A gap exists between what professional accountants understand today and what they need to understand to survive and thrive during the artificial intelligence revolution and the resulting impact on accounting, reporting, auditing, and analysis as we move to an increasingly digital environment.
- This gap in understanding capabilities of technologies such as artificial intelligence makes it difficult for professional accountants to grasp the changes that are not only inevitable but are imminent.
- Undergraduate and graduate college curriculums need to be updated for these changes. Continuing professional education offered likewise needs to be updated.
- The “learn to code” hysteria is not the right answer. Having professional accountants become more proficient with information technology or computer skills is likewise not the right answer for most professional accountants.
- The key to surviving and thriving during the artificial intelligence revolution and in its aftermath is computer empathy. Another term for this is computational thinking.
- The answer is to gain a better understanding of logic. Computers work per the rules of mathematics. Mathematics works per the rules of logic. The primary gap that needs to be filled is for professional accountants to gain a sound understanding of how computers actually work and the skills necessary to leverage these very useful tools of accounting, reporting, auditing, and analysis in a digital environment appropriately.
- Business professionals have an innate understanding of logic. All that they need to do is to supplement this informal natural understanding of logic into a slightly more formal understanding which will enable business professionals to have effective conversations with technical professionals related to employing technology to perform work. Closing this knowledge gap sooner rather than later can help position professional accountants as leaders for the digital age of accounting, reporting, auditing, and analysis.

¹ Charles Hoffman, *Getting Ready for the Digital Age of Accounting, Reporting and Auditing: a Guide for Professional Accountants*, <http://xbrlsite.azurewebsites.net/2017/Library/GettingReadyForTheDigitalAgeOfAccounting.pdf>

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Computers seem to perform magic. How computers do what they do tends to be a mystery to many people. But computers are really simply machines that follow very specific instructions to get work done. Skilled craftsmen who wield their tools effectively which includes providing the appropriate machine-readable instructions enable these machines to perform in mysterious ways and provide the users of these tools with what seems to be magic.

Psychology Today defines empathy² as “the experience of understanding another person’s condition from their perspective”. Borrowing from that definition and modifying it slightly, think of **computer empathy** as the experience of understanding how computer software works from the perspective of the computer. Computer empathy is about understanding how a computer works so that you can better understand that tool and how to employ that tool in your craft to perform useful work reliably, repeatedly, predictably, and safely.

Computer empathy is about demystifying accounting, reporting, auditing, and analysis in a digital environment. No magic; no metaphysics. Engineering.

This document provides a framework and principles to think about computers in a deliberate and conscious manner.

Another term for this is *computational thinking*³. **Computational thinking** is a thought process involved in formulating problems and their solutions so that the solutions are represented in a logical, clear, and systematic form that can be explained to and effectively carried out by a computer or human.

Engineering is the application of a systematic, disciplined, quantifiable, methodical, rigorous approach to the development, operation, and maintenance of something. A kluge is a term from the engineering and computer science world that refers to something that is convoluted and messy but gets the job done. Elegance is the quality of being pleasingly ingenious, simple, neat. Elegance is about beating down complexity. Creating something complex is easy. Creating something simple and elegant is hard work. This document helps those that choose to be master craftsmen and desire to harness the power of the computers in the performance of their craft.

Changes Caused by Artificial Intelligence

The institution of accountancy is ripe for change. A perfect storm has formed that has enabled changes that will significantly change not only the institution of accountancy, but almost everything. Old approaches can be replaced by new more efficient and effective approaches.

² Psychology Today, *Empathy Basics*, <https://www.psychologytoday.com/basics/empathy>

³ Center for Computational Thinking, Carnegie Mellon University, <https://www.cs.cmu.edu/~CompThink/>

To understand this, professional accountants need to understand that a paradigm shift has occurred; old maps will not work to understand the coming changes. Work tasks will be transformed.

But to employ artificial intelligence appropriately the opportunities and the risks need to be consciously understood⁴.

A Perfect Storm

Accounting, which has existed for 10,000 years⁵, even before the creation of formal number systems, is constantly evolving. The institution of accountancy is about to go through another significant phase in its evolution process. Professional accountants who adapt will thrive.

Professional accountants, whose jobs seem secure, are perhaps feeling threatened by technology. This threat by artificial intelligence and other technologies, perhaps, generates fear, uncertainty, doubt and maybe even dread within many different job categories. However, as long as professional accountants are willing and able to adapt to these changes, most professional accountants should be well-positioned to create more value than ever, augmented by machines that they understand how to work with and leverage.

A perfect storm has formed which is driving change not only for the institution of accountancy; rather this change will impact virtually every institution, every organization, every person, and every process. Universal connectivity enabled by the internet brings everything together into one interconnected network of information. Powerful computer hardware that can run powerful software provides almost limitless possibilities. Maturing technologies and techniques such as artificial intelligence that offer increased capabilities for computers to solve problems. Mature and standardized business domain business logic, such as US GAAP and International Financial Reporting Standards (IFRS) that enable precise communication of very complex information. Machine-readable knowledge such as the US GAAP and IFRS XBRL Taxonomies that is useable by both machines and by humans to precisely understand complex knowledge. Individually these things might not seem like much; but combined together correctly and foundational⁶ changes will result.

Business Professionals Augmented by Machines

Some call it *The Digital Industrial Revolution*⁷. Others refer to the changes as *Industry 4.0*^{8,9}. And others call it the *Artificial Intelligence Revolution*^{10,11}. You might have heard other terms to

⁴ Peter Haas, *The Real Reason to be Afraid of Artificial Intelligence*, https://www.youtube.com/watch?v=TRzBk_KulaM

⁵ Wikipedia, *History of Accounting*, retrieved June 10, 2017, https://en.wikipedia.org/wiki/History_of_accounting

⁶ *Incremental Disruptive, and Foundational Technologies*, <http://xbrl.squarespace.com/journal/2018/10/28/incremental-disruptive-and-foundational-technologies.html>

⁷ National Public Radio, TED Radio Hour, *The Industrial Revolution*, retrieved May 4, 2017, <https://www.npr.org/player/embed/524731597/524879464>

describe the rapid technology changes that are occurring around the world. It does not really matter what you call it. What is important to understand is that work will be performed differently in the very near future. Changes will occur and the changes will be significant.

Professional accountants and other business professionals will team with computers more to get work done. This teaming is natural. This teaming is much like how professional accountants use calculators to augment their capabilities to perform mathematics more efficiently.

The Economist predicts¹² that 94% of accounting jobs will be replaced by computers over the next 20 years. That percentage is 98% for accounting clerks, audit clerks, and bookkeepers. While predictions may, perhaps, be overstated; change to some degree is not only inevitable, that change is imminent.

Techno-optimists tend to overstate what change will bring. Overzealous predictions are not helpful. On the other hand, business professionals tend to understate what change will bring, minimizing productivity increases that might be possible because, perhaps, they don't particularly care for change or because they really don't understand the technology enough to make accurate predictions.

This paper strives to cut through the rhetoric and provide professional accountants with the information they need to understand, evaluate, and leverage the maturing technologies which will surely have a very significant impact on every business professional and every company in business today. This paper provides details about specific skills professional accountants need to acquire in order to even understand the changes that are inevitable and imminent related to accounting, reporting, auditing, and analysis.

Antiquated Old School Accounting, Reporting, Auditing, and Analysis Practices and Procedures

The following are descriptions of the current state of accounting, reporting, auditing and analysis processes that are generally used today.

Accounting

⁸ Cornelius Baur and Dominik Wee, McKinsey & Company, *Manufacturing's Next Act*, <http://www.mckinsey.com/business-functions/operations/our-insights/manufacturings-next-act>

⁹ PriceWaterhouseCoopers, *Industry 4.0: Building the digital enterprise*, <http://www.pwc.com/gx/en/industries/industry-4.0.html>

¹⁰ Steven Levy, *Wired*, *The AI Revolution is On*, https://www.wired.com/2010/12/ff_ai_essay_airevolution/

¹¹ *Wired*, The President in Conversation With MIT's Joi Ito and WIRED's Scott Dadich, *Barack Obama, Neural Nets, Self-driving Cars, and the Future of the World*, <https://www.wired.com/2016/10/president-obama-mit-joi-ito-interview/>

¹² *The Economist*, Jan 18, 2014, *The future of jobs: The onrushing wave*, <http://www.economist.com/news/briefing/21594264-previous-technological-innovation-has-always-delivered-more-long-run-employment-not-less>

Blackline¹³, an accounting process automation software vendor, describes the accounting process:

“Most finance organizations still cope with a rigid accounting system. Their accountants do basic bookkeeping until the end of the accounting period, with their noses kept to the grindstone to close the books accurately and on time. Every company accountant will attest to the hardships of their period-end tasks, working long days into late evenings, and eating lunch and dinner at their desks. Exhausted to the point of burnout, they’re more likely to make errors at a time when data accuracy counts most.”

“While technology has revolutionized the modern business world, the record-to-report process has remained unchanged since the days of Lotus Notes and 18 inch paper. ERP systems and spreadsheets are still the only traditional tools available to today’s accounting and finance professionals, and while this is accepted by many as **“the way it’s always been done,”** it results in a time-consuming, inefficient process that is fraught with risk. Talented accountants are spending most of their time performing rote tasks that take them away from value-adding activities, like the enhanced analysis and strategy that drew them to this profession in the first place.”

“With transaction volumes increasing exponentially, however, throwing more resources at the problem is no longer an effective or sustainable solution. And with staff mired in mundane, low-value tasks, engagement is on the decline at a time when focusing on analysis and increasing productivity is more important than ever.”

This is not a unique description of the accounting process.

Reporting

By all accounts, the process of creating an external financial report is an extremely inefficient and outdated process. Here is a list of how some describe that process:

- The CFA Institute calls for “...greater efficiencies within the current inefficient system” [of creating financial reports]¹⁴.
- The consultancy Gartner points out, “...average Fortune 1000 company used more than 800 spreadsheets to prepare its financial statements”¹⁵
- Ventana Research says, “...for larger companies, assembling the periodic external reports typically is an inefficient and error-prone process.”¹⁶

¹³ Blackline, Accounting Process Automation, <https://www.blackline.com/accounting-process-automation>

¹⁴ CFA Institute, *DATA AND TECHNOLOGY: TRANSFORMING THE FINANCIAL INFORMATION LANDSCAPE*, June 2016, <http://www.cfapubs.org/doi/pdf/10.2469/ccb.v2016.n7.1>

¹⁵ Nigel Rayner, Neil Chandler, *XBRL Will Enhance Corporate Disclosure and Corporate Performance Management*, April 23, 2008, <http://unstats.un.org/unsd/nationalaccount/workshops/2008/newyork/IG22.PDF>

- PriceWaterhouseCoopers points out, "...old school manual processes..." and "commonly cut and pasted, rekeyed, or manually transferred into word processing and spreadsheet applications used for report assembly and review process steps"¹⁷

While the process of creating an external financial report might not seem inefficient when being measured against current practices, procedures, processes, and mentalities of those thinking that the way financial reports are created today is “the only way” to create such reports; and while it is hard to measure the effectiveness and efficiency of new practices, procedures, and processes because they don’t exist yet; when new practices, procedures, and processes do exist the increase in productivity will be measurable and clear, and they will be substantial.

Auditing

In their paper, *Imagineering Audit 4.0*¹⁸, Jun Dai and Miklos A. Vasarhelyi point out that for a number of reasons auditors are not keeping up with advancing technology:

“Traditional manual audits (Audit 1.0) have existed for centuries fulfilling many needs. Although the IT audit (Audit 2.0) emerged in the 1970s, and most all businesses are currently computer based, only about 15 percent of auditors are IT enabled. This delay of IT adoption can be partially attributed to the conservatism and rigidity of the profession, as well as the calcifying effect of increasingly obsolete regulation, but also to the lack of quality tools that would allow traditional auditors to automate the functions that they currently perform manually.”

The territory has changed. Robert Kugel of Ventana Research coined the term “**robotic finance**” and explains his vision in an excellent article, *Welcome to the Age of Robotic Finance*¹⁹. In addition, Mr. Kugel elaborates on this vision in more detail in a very informative 35 minute webinar²⁰.

Analysis

Chances are that you use spreadsheets for data analysis and analytics in some capacity. This is with good reason: spreadsheets are easy to use for simple data analysis tasks. But spreadsheets have

¹⁶ Robert Kugel, Ventana Research: Making XBRL Reporting Easy, February 13, 2009, <http://businessfinancemag.com/technology/ventana-research-making-xbrl-reporting-easy>

¹⁷ Mike Willis, PriceWaterhouseCoopers, Disclosure management: Streamlining the Last Mile, March 2012, <https://www.pwc.com/gx/en/xbrl/pdf/pwc-streamlining-last-mile-report.pdf>

¹⁸ Jun Dai and Miklos A. Vasarhelyi (2016) *Imagineering Audit 4.0*. Journal of Emerging Technologies in Accounting: Spring 2016, Vol. 13, No. 1, pp. 1-15.; <http://aaajournals.org/doi/abs/10.2308/jeta-10494>

¹⁹ Ventana Research, Robert Kugel, *Welcome to the Age of Robotic Finance*, <https://robertkugel.ventanaresearch.com/welcome-to-the-age-of-robotic-finance>

²⁰ Ventana Research, Robert Kugel, *The digital transformation of finance departments is finally within reach*, https://www.ventanaresearch.com/webinar/office_of_finance/robotic_finance

limitations. Relying on spreadsheets for sophisticated data analysis and business decision-support can be inefficient and could expose you to significant risk. So, while spreadsheets such as Excel are excellent tools; spreadsheets are not a panacea and tend to be inappropriate in many cases²¹. Professional accountants will need to learn increasingly sophisticated data analysis and business analytics tools.

Accounting, Reporting, Auditing, and Analysis in a Digital Environment

A general purpose financial report is a high-fidelity, high-resolution, high-quality information exchange mechanism. That mechanism has historically used the media of "paper". Over the past 50 years or so, that paper-based mechanism has given way to a new mechanism, "e-paper". By "e-paper" I mean PDF documents, HTML documents, Word documents and such. XBRL is a new media²², a new mechanism for creating a general purpose financial report. XBRL is a high-fidelity, high-resolution information exchange media that allows for the creation of high-quality financial reports.

In an article, *Surety Data Standards: Is Manual Data Entry Dead?*²³, NASBP says "We're hammering the final nail in the coffin of manual data entry... more to come." NASBP says, "The gruesome (and grueling) days of painful re-keying of data may be coming to an end. Could data standards be the magic bullet?" The article points out:

"In 2017, The Hartford successfully brought standardized WIP reports into their internal financial management system, *reducing WIP report processing from 20 minutes to 3 seconds*. And earlier this year, two more software providers enabled their applications to prepare XBRL-formatted WIPs. Crowe LLP introduced a unique application to automatically build XBRL-formatted WIP reports along with other financials."

The many painful, monotonous, grueling, onerous, and even gruesome tasks related to accounting, reporting, auditing, and analysis will be made easier by professional accountants augmenting their capabilities similar to how calculators help you improve your math capabilities.

"The map is not the territory"

Again, the territory has changed. A paradigm shift has occurred. But accountants and auditors are using the same map, or mental model, that they have used for 50 years to think about accounting and auditing. That map is outdated. As Stephen Covey pointed out in his seminal book, *The Seven Habits of Highly Effective People*²⁴, it is like being in Chicago and having a map of Detroit and trying to use that map of Detroit to make it to a specific location in Chicago. It is very frustrating and tends to be rather ineffective.

²¹ Bernard Marr, *Excel Reporting: 5 Reasons Why It Is Bad For Business*, June 16, 2016, Forbes, <https://www.forbes.com/sites/bernardmarr/2016/06/16/spreadsheet-reporting-5-reasons-why-it-is-bad-for-business/>

²² Understanding that XBRL is a Knowledge Media, <http://xbrl.squarespace.com/journal/2017/1/16/understanding-that-xbrl-is-a-knowledge-media.html>

²³ NASBP, *Surety Data Standards: Is Manual Data Entry Dead?*, <https://www.nasbp.org/pipeline/morearticles/manual-entry-end>

²⁴ Stephen R. Covey, *The Seven Habits of Highly Effective People*, page 23.

In order to understand the changes that are occurring we first need to help professional accountants update their mental map for thinking about computers, accounting, reporting, auditing, and analysis. The paradigm has already shifted, the world has changed and there is no going back.

Enhancing your “Digital Skills”

Professional accountants will need to improve their “digital skills” in order to be efficient at accounting, reporting, auditing, and analysis in a digital environment.

Allan Grody, in his article, *Banking could use more leaders with STEM skills*²⁵, points out the importance for those in leadership positions to become conversant in the important details about the technologies they will inevitably employ:

“It’s important that those at the front end of this information — legislators, regulators and financial executives — become conversant in these skills, lest the back-end stumbles over meaning and intent and, in some cases, declares it impossible to be implemented.”

While those in leadership positions don’t need to have a deep understanding of math or computer programming; they do need to have the capacity to ask good probing questions and be able to have a meaningful conversation about the digital construction materials those that they lead are working with. Leaders should be able to question and challenge how their direct reports are overseeing the implementation digital tools that will be used to perform work using algorithms, metadata, conceptual models, and other tools, techniques, and processes.

Don’t be an information barbarian²⁶. Leaders have a responsibility to acquire these “digital skills” that allow them to understand how accounting, reporting, auditing, and analysis work in a digital environment; ultimately industry will hold these leaders to a higher standard.

STEM (science, technology, engineering, and math) is a very general description. “Learn to code” is a hysteria, which likewise is a very general description and it is simply misguided to think that everyone should be a programmer. Understanding the specifics of how computers work will help you understand how to employ these very useful tools. This document strives to provide you with details of what you need to learn with enough information to help you

²⁵ Allan D. Grody, *Banking could use more leaders with STEM skills*, American Banker, July 9, 2018, <https://www.americanbanker.com/opinion/banking-could-use-more-leaders-with-stem-skills>

²⁶ Information Barbarian, <http://xbri.squarespace.com/journal/2017/1/27/information-barbarians.html>

understand why you should learn it. This document should provide you with a good framework²⁷ and principles²⁸ for working in a digital environment.

Rearranging Abstract Symbols

Computers sometimes seem to perform magic. But computers are really simply machines that follow very specific instructions. Skilled craftsmen, who wield their tools effectively, providing the correct machine-readable instructions, create what seems to be magic.

In his book *Saving Capitalism*²⁹, Robert Reich describes three categories that all modern work/jobs fit into:

- **Routine production services** which entails repetitive tasks,
- **In-person services** where you physically have to be there because human touch was essential to the tasks,
- **Symbolic-analytic services** which include problem solving, problem identification, and strategic thinking that go into the manipulation of symbols (data, words, oral and visual representations).

In describing the third category, symbolic-analytic services, Mr. Reich elaborates:

“In essence this work is to rearrange abstract symbols using a variety of analytic and creative tools - mathematical algorithms, legal arguments, financial gimmicks, scientific principles, powerful words and phrases, visual patterns, psychological insights, and other techniques for solving conceptual puzzles. Such manipulations improve efficiency- accomplishing tasks more accurately and quickly-or they better entertain, amuse, inform, or fascinate the human mind.”

Many tasks in accounting, reporting, auditing, and analysis are related to symbolic-analytic services.

Shelly Palmer breaks work tasks down in another way³⁰. He points out that almost every human job requires us to perform some combination of the following four basic types of tasks:

- Manual repetitive (predictable)
- Manual nonrepetitive (not predictable)
- Cognitive repetitive (predictable)

²⁷ *Understanding the Need for a Framework and Theory*, <http://xbri.squarespace.com/journal/2015/9/20/understanding-the-need-for-a-framework-and-theory.html>

²⁸ *XBRL-based Digital Financial Reporting Principles*, <http://xbri.squarespace.com/digital-financial-reporting-pr/>

²⁹ Robert B. Reich, *Saving Capitalism*, Alfred A. Knopf, page 204-206, (see a summary from *The Work of Nations* here http://www.oss.net/dynamaster/file_archive/040320/e8eb8748abfe77204a145d5fbcc892fb/QSS1993-01-37.pdf#page=6)

³⁰ Shelly Palmer, *The 5 Jobs Robots Will Take Last*, <https://www.linkedin.com/pulse/5-jobs-robots-take-last-shelly-palmer>

- Cognitive nonrepetitive (not predictable)

Manual involves using one's hands or physical action to perform work. *Cognitive* involves using one's brain or mental action or a mental process of acquiring knowledge/understanding through thought, experience, use of the senses, or intuition. Predictable manual or cognitive tasks can be automated. Unpredictable manual or cognitive tasks cannot be automated. He gives the example of an assembly line worker that performs mostly manual repetitive tasks which, depending on complexity and a cost/benefit analysis, can be automated. On the other hand, a CEO of a major multinational conglomerate performs mostly cognitive nonrepetitive tasks which are much harder to automate.

A Brief History of Ledgers

Between 5,000 and 10,000 years ago farmers in Mesopotamia, where agriculture was born, used physical object to count crops and animals³¹. The distinction between types of crops or animals was made by using different types and shapes of objects. Then, in about 3200 BC, around 5,000 years ago, the first spreadsheet was invented.

These farmers began documenting information using clay tablets in the earliest form of human writing ever discovered called Cuneiform. They partitioned their clay tablet into rows, columns, and cells. These farmers used single-entry accounting. The spreadsheet below documents an account of barley distribution³²:



³¹ Denise Schmandt-Bessersat, *On the origins of writing*, YouTube.com, <https://www.youtube.com/watch?v=kidWY-pJFb0>

³² Metropolitan Museum, Proto-Cuneiform tablet with seal impressions: administrative account of barley distribution with cylinder seal impression of a male figure, hunting dogs, and boars, <https://www.metmuseum.org/art/collection/search/329081>

In 1211 AD a bank in Florence was the first documented use of double-entry accounting³³. Between 1299 AD and 1300 AD double-entry accounting came of age. In 1494 AD during the Renaissance, Venetian mathematician and Franciscan friar Luca Pacioli³⁴ published a book, *Summa de arithmetica, geometria. Proportioni et proportionalita* (*Sum of Arithmetic, Geometry, Proportion and Proportionality*)³⁵. That book documented an approach to accounting now called double-entry bookkeeping and recommended that others use this approach. The approach allowed for better error detection and the ability to differentiate unintended errors from fraud. Accountants adopted that new approach.

Accountants have a special name for the spreadsheets, or tables, that these farmers invented. Accountants call these *ledgers*³⁶. A ledger is simply a place where you record information such as transactions. There are a number of different types of ledgers that accountants use. General ledgers³⁷ tend to be double-entry and capture numeric information. Special ledgers, or subsidiary ledgers, are also generally double-entry and are basically special purpose ledgers for recording specific classes of transactions. For example, a sales ledger is used to record sales transactions of an economic entity and a purchases ledger is used to record purchases of an economic entity.

There are many other types of ledgers used that are related to or unrelated to keeping the accounts of an economic entity. Some of these ledgers are public ledgers, others are private ledgers³⁸. Most of these tend to be single-entry. Here are some examples of ledgers:

- Property titles
- Chain of custody
- Building permits
- Delivery records
- Digital rights management
- Insurance
- Proof of authenticity
- Anti-money laundering

Most recently there are discussions about immutable digital distributed ledgers using cryptographic technologies³⁹. The notion of triple-entry accounting⁴⁰ was first articulated in

³³ Geoffrky Alan Lee, *The Development of Italian Bookkeeping 1211–1300*, Wiley, <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1467-6281.1973.tb00183.x>

³⁴ Wikipedia, Luca Pacioli, https://en.wikipedia.org/wiki/Luca_Pacioli

³⁵ Wikipedia, Summa de arithmetica, https://en.wikipedia.org/wiki/Summa_de_arithmetica

³⁶ Wikipedia, Ledger, <https://en.wikipedia.org/wiki/Ledger>

³⁷ Wikipedia, General Ledger, https://en.wikipedia.org/wiki/General_ledger

³⁸ InterChainZ, Sharing Ledgers for Sharing Economies, YouTube.com, <https://www.youtube.com/watch?v=Hwhigpr4720>

³⁹ Understanding Digital Distributed Ledgers, <http://xbrl.squarespace.com/journal/2015/12/3/understanding-digital-distributed-ledgers.html>

about 1998⁴¹. The blockchain community and others⁴² seem to have borrowed that term and changed what it means⁴³. Others can debate who owns the term “triple-entry”. Some say that triple-entry is the most important invention in 500 years⁴⁴. People are already experimenting with triple-entry accounting systems⁴⁵. The way the blockchain people tend to use the term “triple-entry” is taking two double-entry systems and putting a publically available link between the two double-entry systems documenting that the transactions go together. I don’t know if the idea has been proven, but intuitively it seems like a very useful tool.

Ledgers, Tables, Spreadsheets, and Financial Reports are Abstract Ideas

A balance sheet and an income statement are abstract ideas invented by humans. The stuff in the report is symbols. Rearranging abstract symbols, such as the creation of such financial reports, can be achieved using human-based processes and tools or using machine-based processes and tools. Likely a combination of human and machine based processes will be employed in the future; humans performing the tasks they do best, computers performing tasks that they do best.

There is nothing natural about the ledger⁴⁶. A ledger was an invention of man. The columns in a ledger are abstractions. The numbers and other information that go into the columns are symbols. The ledger is a useful idea. Professional accountants have used ledgers for many, many years.

A table⁴⁷ is likewise a useful idea, an abstraction. A table has rows. A table has columns. A table has cells which are the intersection of a row and a column. So actually, do you realize that a table can have way, way more than rows, columns, and cells? What about groups of rows. And how about groups of columns. What about a row that spans more than one column. You might not have known that there is actually a standard specification which describes tables, the CALS Table Model⁴⁸. The spreadsheet⁴⁹ is likewise an abstract idea.

⁴⁰ Marcell Nimfuehr, BlockchainTech: Can Triple Entry Accounting Save the World?, <https://blog.goodaudience.com/blockchaintech-can-triple-entry-accounting-save-the-world-896092da4694>

⁴¹ Yuji Ijiri, Interview about Triple Entry Bookkeeping, YouTube.com, <https://www.youtube.com/watch?v=7YE8IWl3tAA>

⁴² Ian Grigg, Triple Entry Accounting, http://iang.org/papers/triple_entry.html

⁴³ Ben Taylor, Triple-Entry Accounting And Blockchain: A Common Misconception, Forbes, <https://www.forbes.com/sites/forbesfinancecouncil/2017/11/28/triple-entry-accounting-and-blockchain-a-common-misconception/#b180cd0190f9>

⁴⁴ Daniel Jeffries, *Why Everyone Missed the Most Important Invention in the Last 500 Years*, <https://hackernoon.com/why-everyone-missed-the-most-important-invention-in-the-last-500-years-c90b0151c169>

⁴⁵ YouTube.com, DEVCON1: Balanc3 - Triple Entry Accounting, <https://www.youtube.com/watch?v=NEYTypMoQv0> (if you watch the video, note the reference to the link to the XBRL document about 4 minutes into the video)

⁴⁶ Wikipedia, *Ledger*, retrieved August 8, 2017, <https://en.wikipedia.org/wiki/Ledger>

⁴⁷ Wikipedia, *Table (Information)*, retrieved August 8, 2017, [https://en.wikipedia.org/wiki/Table_\(information\)](https://en.wikipedia.org/wiki/Table_(information))

⁴⁸ Wikipedia, *CALS Table Model*, retrieved August 8, 2017, https://en.wikipedia.org/wiki/CALS_Table_Model

Why would someone create a written specification for a table? Well, that is how you make it so machines can read a table and that tables created on one machine can also be read on another different machine. A specification explains how to create tables consistently, the patterns necessary to get a machine to be able to work with an electronic version of a table.

Harnessing the power of the dumb beasts

Computers are dumb beasts. However, if a skilled craftsmen uses the right tools; a computer can seem to perform magic.

Computers are General Purpose Machines

Professional accountants need to understand some fundamentals about how these machines that we call computers work and about their true capabilities.

Computers are machines. The first mechanical computers, called tabulating machines⁵⁰, were created in the 1900s. Since then the effectiveness and efficiency of those machines have improved by orders of magnitude. But fundamentally the machines we use today are no different than mechanical tabulating machines of the past. The key word here is machine.

A computer has two distinct parts: hardware and software. What makes computers such a unique and compelling tool is not the hardware, although that is very important; it is the software. Most machines tend to do one thing. Computers are extremely useful tools because by changing only software you can make the computer do different things. And because software is flexible, you can make a computer perform many, many different tasks. Computers are general purpose machines.

Computers are Dumb Beasts

Important to understanding how to get computers to do what you want is understanding how computers actually work. The strengths of computers and the obstacles that get in the way of using computers were summarized well by Andrew D. Spear⁵¹; here is his list with some modifications made by me:

Fundamental strengths/capabilities of computers:

- **store** tremendous amounts of information reliably and efficiently
- **retrieve** tremendous amounts of information reliably and efficiently

⁴⁹ Wikipedia, *Spreadsheet*, retrieved August 8, 2017, <https://en.wikipedia.org/wiki/Spreadsheet>

⁵⁰ Wikipedia, *Tabulating Machine*, https://en.wikipedia.org/wiki/Tabulating_machine

⁵¹ Andrew D. Spear in his document, *Ontology for the Twenty First Century: An Introduction with Recommendations*, page 4

- **process** stored information reliably and efficiently, mechanically repeating the same process over and over
- make information **instantly accessible** to individuals and more importantly other machine-based processes anywhere on the planet in real time

Major obstacles to harnessing the power of computers:

- **business professional idiosyncrasies**; different business professionals use different terminologies to refer to exactly the same thing
- **information technology idiosyncrasies**; information technology professionals use different technology options, techniques, and formats to encode and store exactly the same information
- **inconsistent domain understanding** of and technology's limitations in expressing interconnections within a domain of knowledge
- **computers are dumb beasts**; computers don't understand themselves, the programs they run, or the information that they work with

Keep in mind that the information business professionals are trying to store and make use of is becoming more complex than what they have been storing in relational databases or spreadsheets for the past 50 years. For example, a financial report is complex information that is very difficult to store in a relational database and query across millions of such reports efficiently.

Difference Between Data and Information

There are specific differences between data, information, knowledge, and wisdom⁵²:

- **Data**: The basic compound for intelligence is data. Data are measures, observations, symbols, phenomenon, utterances, and other such representations of the world around us presented as external signals and picked up by various sensory instruments and organs. Simplified: data is raw facts and numbers.
- **Information**: Information is produced by assigning relevant meaning related to the context of the data to the data. Simplified: information is data in context.
- **Knowledge**: Knowledge is the understanding or interpretation, a justifiable true belief, of information and approach to act upon the information in the mind of the perceiver. Simplified: knowledge is the interpretation of information.
- **Wisdom** (or Intelligence or Understanding): Wisdom embodies awareness, insight, moral judgments, and principles to construct new knowledge and improve upon existing understanding. Simplified: wisdom is the creation of new knowledge.

⁵² Wikipedia, *DIKW Pyramid*, retrieved February 24, 2016, https://en.wikipedia.org/wiki/DIKW_Pyramid

Information Storage Structures

People tend to agree that there are three primary formats for representing structured and information:

- **Table-type format**⁵³: a table is an arrangement of data in rows and columns which intersect to form cells; relational databases, CSV, spreadsheets, or tabular-type representations which allow only one level of hierarchy within each table; but more complex hierarchies can be constructed by relating tables.
- **Tree-type format**⁵⁴: a tree is a data structure made up of nodes and edges without having any cycles; a tree is a special case of a graph; XML, XBRL (using tuples), JSON and other tree-hierarchy-type information which allow for the expression of one hierarchy; hierarchical type databases, object type databases (Note that a tree is a special case of graph.)
- **Graph-type format**⁵⁵: a graph is a data structure made up of nodes and edges; RDF, EAV, XBRL (using dimensions), JSON-LD and other open schema-type or graph-type representations which are more graph-oriented or network-oriented (many-to-many) and allow for dynamically creating virtually any number of hierarchies; supports the notion of cycles; very flexible; network type databases; RDF triple stores.

Most people understand tables. Trees and graphs are typically less understood data structures.

Directed acyclic graphs⁵⁶ are the "sweet spot" where you get all the advantages of trees, some of the advantages of graphs, but without the catastrophic consequences that can be caused by certain types of cycles. For example, spreadsheets and tables can be modeled as a directed acyclic graph.

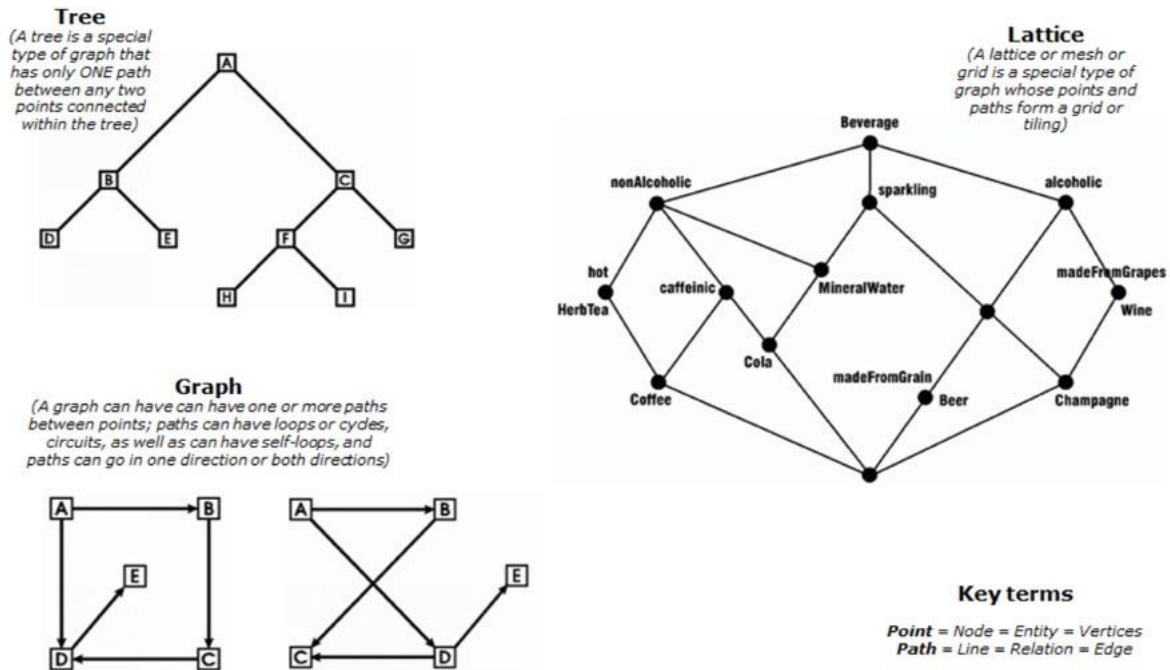
This graphic helps the reader understand graphs and the difference between a graph and a tree:

⁵³ Wikipedia, *Table (Information)*, retrieved March 10, 2018, [https://en.wikipedia.org/wiki/Table_\(information\)](https://en.wikipedia.org/wiki/Table_(information))

⁵⁴ Wikipedia, *Tree (Data Structure)*, retrieved March 10, 2018, [https://en.wikipedia.org/wiki/Tree_\(data_structure\)](https://en.wikipedia.org/wiki/Tree_(data_structure))

⁵⁵ Wikipedia, *Graph theory*, retrieved March 10, 2018, https://en.wikipedia.org/wiki/Graph_theory

⁵⁶ Wikipedia, *Directed acyclic graphs*, retrieved May 17, 2018, https://en.wikipedia.org/wiki/Directed_acyclic_graph



- **Graph:** A graph can have one or more paths between points; paths can have loops or cycles, circuits, as well as can have self-loops, and paths can go in one direction or both directions.
- **Tree:** A tree is a special type of graph that has only one path between any two points connected within the tree.
- **Lattice:** A lattice or mesh or grid is a special type of graph whose points and paths form a grid or tiling.

Things, Not Strings

Jo Stichbury, in his article *WTF is a knowledge graph*⁵⁷, does a great job of explaining what a knowledge graph is and why it is useful. Here are the key points as I see them related to knowledge graphs:

- A knowledge graph is a graph rather than a table, or a tree. Basically, a graph is a powerful and flexible way to store knowledge.
- A knowledge graph is self-describing; basically there is logic that explains the meaning being conveyed within the knowledge graph.
- It is not simply about the knowledge graph itself but more about the combination of the knowledge graph and tools to work with the information within the knowledge graph. The object is to make use of the information in the knowledge graph.

⁵⁷ Jo Stichbury, *WTF is a knowledge graph*, <https://hackernoon.com/wtf-is-a-knowledge-graph-a16603a1a25f>

If you want to understand more about knowledge graphs, see this paper *Towards a Definition of Knowledge Graphs*⁵⁸.

So why are knowledge graphs plus the tools for using information from a knowledge graph so important? Using the right tools, a skilled craftsman can easily produce works that are useful, elegant, and durable. Most tools today are not the right tools. They are clumsy, ugly, barely functional software that does not remotely meet their needs when it comes to working with structured information such as XBRL. This is in part because many software engineers think they are working with strings and data when they really need to think in terms of semantics or “things” and information.

Standards Minimize but Don’t Eliminate Idiosyncrasies

Standards can be created to minimize the idiosyncrasies of both business professionals and information technology professionals. A complete discussion of how standards can be useful in minimizing idiosyncrasies is beyond the scope of this document. But I will point out that standard shipping containers, uniform product codes, standard electrical outlets, standard measurement tools, and other such standards are quite useful⁵⁹.

That said, people will be people. Fads, trends, personal preferences, fallacies, misinformation and other dynamics impact people’s choices. They always have and they always will. As such, it is highly unlikely that you can get everyone to agree on something ever. So, multiple options can be a very good thing. Whether someone’s needs are truly different or perceived to be different; having multiple alternatives rather than one single alternative can be a very good thing.

Except, when it is not. Agreeing to agree is great when it is possible. There is truly art and science involved in properly balancing something to get that something to work in equilibrium.

The important thing to keep in mind is that standards do exist and when employed appropriately they can save a lot of time and money.

Role of Standards

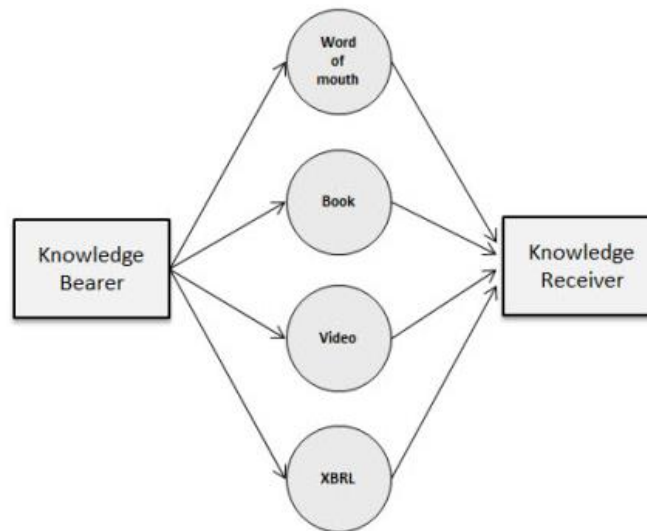
The role of standards tends to be misunderstood and often under appreciated. These standards make things easier. There are many different standards such as intermodal shipping containers⁶⁰, universal product codes, the metric system, JPEG photo format, MPEG audio

⁵⁸ Lisa Ehrlinger and Wolfram Wöß, *Towards a Definition of Knowledge Graphs*, <http://ceur-ws.org/Vol-1695/paper4.pdf>

⁵⁹ Standards Make Markets, <http://xbrl.squarespace.com/journal/2018/12/22/standards-make-markets.html>

⁶⁰ Wikipedia, *Intermodal shipping container*, https://en.wikipedia.org/wiki/Intermodal_container#Specifications

format, etc. Simply put, standards make things easier for users. XBRL is a global standard knowledge media⁶¹.



In order to make use of a knowledge media effectively, the following three conditions must be satisfied:

1. **Easy for knowledge bearer to represent information:** The effort and difficulty required for the knowledge bearer to successfully formulate the knowledge in the medium must be as low as possible.
2. **Clear, consistent meaning:** The meaning conveyed by the knowledge bearer to the knowledge receiver must be clear and easily followed by human beings and be consistent between different software applications. The result cannot be a "black box" or a guessing game and users of the information should not be able to derive different knowledge simply by using a different software application.
3. **High-quality information representation:** The form in which the knowledge is represented to the receiver must be as good as possible. The quality must be high whether the knowledge receiver is a human-being or an automated machine-based process. Sigma level 6⁶² is a good benchmark, 99.99966% accuracy.

Classification Systems

Things in the world are defined by their relations to one another; these explicit relations matter in creating logical definitions. A classification system is a logical grouping of something based

⁶¹ Understanding that XBRL is a Knowledge Media, <http://xbrl.squarespace.com/journal/2017/1/16/understanding-that-xbrl-is-a-knowledge-media.html>

⁶² Wikipedia, *Sigma Levels*, retrieved February 24, 2016, https://en.wikipedia.org/wiki/Six_Sigma#Sigma_levels

on some similarity or criteria. A classification system is a communications tool. A classification system structures information. A classification system can be informal or formal, more rigorously or less rigorously created, readable/usable by computers, or not. A classification system can be a controlled vocabulary. Classification systems can be categorized as follows:

- A **dictionary** or list is a classification system that tends to provide descriptions without much, or any, structure. Dictionaries or lists simply provide a flat inventory of terms with no relations expressed between the terms. (But even a dictionary classifies terms into noun, verb, adverb, etc.)
- A **taxonomy** is a classification system which tends to provide descriptions and a limited amount of structure generally in the form of one hierarchy into which some list of terms is categorized. Categories are basically sets. A taxonomy is a tree of categories of things with only one relation expressed so terms appear in only one location in a hierarchy of categories. A creator of a taxonomy creates concepts, creates coherent definitions for those concepts, and puts concepts into “buckets” or categories.
- An **ontology** is a classification system which tends to provide descriptions and multiple structures and therefore tends to have more than one hierarchy into which terms are categorized. So an ontology can be thought of as a set of taxonomies. An ontology can express many different types of relations which includes traits/qualities of each term. An ontology is less like a tree and more like a graph⁶³ (network theory). This distinction is very important. The creator of an ontology identifies and establishes models explaining how things in a given ontology are related to one another, the kinds of relationships that exist, the rules of the model. If an ontology provides enough information, it can describe a conceptual model.

Understanding Metadata; Third Order of Order

Putting data into a database is helpful. But if you don’t construct your database correctly you might not have a solid foundation to build upon. While data is important, the real power of computers comes from adding metadata to your data.

David Wenberger's book *Everything Is Miscellaneous*⁶⁴ explains that there are three orders of order:

- **First order of order.** Putting books on shelves is an example the first order of order.

⁶³ Wikipedia, *Network Theory*, retrieved February 24, 2016; https://en.wikipedia.org/wiki/Network_theory

⁶⁴ David Wenberger, *Everything Is Miscellaneous*, https://books.google.com/books?id=KDDwp8zXCQMC&printsec=frontcover&dq=everything+is+miscellaneous&source=bl&ots=LJ55ZUklBG&sig=LMz39YbfNW1siYVT_O3TrryoHnQ&hl=en&ei=pctGTbnINyVksQOpkuTBCg&sa=X&oi=book_result&ct=result&sqi=2#v=onepage&q&f=false

- **Second order of order.** Creating a list of books on the shelves you have is an example of second order of order. This can be done on paper or it can be done in a database. You can use the paper or database list to locate books.
- **Third order of order.** Adding even more information to information is an example of third order of order. Using the book example, classifying books by genre, best sellers, featured books, bargain books, books which one of your friends has read; basically there are countless ways to organize something.

The solution to the information overload problem is to create more information: metadata. Metadata⁶⁵ is simply data that provides information about other data. Machine-readable metadata adds perspective and context to data. People sometimes get into philosophical debates about what is data and what is metadata, but this is to completely miss the point.

This is what you need to know about metadata. Metadata is a good thing. More metadata is better. Standard metadata is even better. An example of metadata is the card catalog of a library. Metadata is generally organized into some sort of classification system. There are three types of metadata⁶⁶:

- **Descriptive:** describes and identifies information
- **Structural:** organizes the types and parts of information and how the parts are related to one another
- **Administrative:** provides other information that helps use some sort of system.

Acquiring a Thick Metadata Layer

A key ingredient to a computer-based system is metadata. The power of a computer system is proportional to the high-quality metadata available. What is not in dispute is the need for a "thick metadata layer" and the benefits of that metadata in terms of getting a computer to be able to perform useful and meaningful work.

But what is sometimes disputed is *how* to most effectively and efficiently get that thick metadata layer. There are two basic approaches to getting this thick metadata layer:

- **Have the computer figure out what the metadata is:** This approach uses artificial intelligence, machine learning, and other high-tech approaches to detecting patterns and figuring out the metadata.

⁶⁵ Wikipedia, *Metadata*, <https://en.wikipedia.org/wiki/Metadata>

⁶⁶ YouTube, *Basics of Metadata*, <https://www.youtube.com/watch?v=0vc6LeVa14>

- **Tell the computer what the metadata is:** This approach leverages business domain experts and knowledge engineers to piece together the metadata so that the metadata becomes available.

Because knowledge acquisition can be slow and tedious, much of the future of computer systems depends on breaking the metadata acquisition bottleneck and in codifying and representing a large knowledge infrastructure. However, this is not an “either/or” question. Both manual and automated knowledge acquisition methods can be used together.

There is a lot of talk about neural networks⁶⁷ enabling things like machine learning⁶⁸ and deep learning⁶⁹. There are two important points that business professionals tend to miss or software vendors creating such software tend to leave out of their sales pitches. First, the amount of training⁷⁰ that is necessary to get a neural network to work correctly. The training process is time consuming, expensive and error prone.

Second, because the automated process is error prone; there are good uses for neural networks figuring out the “thick layer of metadata”, and there are very bad uses. One description of what neural networks are best for is the following:

Neural networks are universal approximators, and they work best if the system you are using them to model *has a high tolerance to error*. One would therefore not be advised to use a neural network to balance one's check book! However they work very well for:

- capturing associations or discovering regularities within a set of patterns;
- where the volume, number of variables or diversity of the data is very great;
- relationships between variables are vaguely understood; or,
- relationships are difficult to describe adequately with conventional approaches.

And so, the probability of a neural network figuring out something like the US GAAP Financial Reporting XBRL Taxonomy is basically zero. However, that said; if such metadata is created and then that human-created metadata is used to then train neural networks the probability that the neural network can create something useful goes up dramatically.

So, again, this is not an “either-or” proposition. This is about using the right tool for the right job and not being misguided by snake oil salesmen who don't have your interest in mind.

⁶⁷ *A Basic Introduction To Neural Networks*, <http://pages.cs.wisc.edu/~bolo/shipyard/neural/local.html>

⁶⁸ Wikipedia, *Machine Learning*, retrieved August 29, 2016, https://en.wikipedia.org/wiki/Machine_learning

⁶⁹ Wikipedia, *Deep Learning*, retrieved August 29, 2016, https://en.wikipedia.org/wiki/Deep_learning

⁷⁰ Hugo Larochelle, et. al., *Deep Learning and Applications in Neural Networks*, <http://www.slideshare.net/hammawan/deep-neural-networks>

Notions, Not Notations

A 19th century mathematician was once said to have quipped that solving a math problem was a question of "notions, not notations." What he meant was that the problem was not the *syntax of the notations*, the problem was the different *ideas people were using* because there was a lack of clear, common understanding between parties related to the ideas.

If we have a shared foundational logical model and a common methodology, we have the chance to automate the exchange and meaningful reuse of information.

Business Rules are Metadata

The Merriam-Webster dictionary defines anarchy⁷¹ as “a situation of confusion and wild behavior in which the people in a country, group, organization, etc., are not controlled by rules or laws.” Business rules prevent information anarchy⁷². Business rules are metadata.

Business rules enable a knowledge bearer to describe information they are providing and verify that the information provided is consistent with that description. Business rules enable a knowledge receiver to understand the description of information provided by the knowledge bearer and likewise verify that the information is consistent with that description.

Business rules guide, control, suggest, or influence behavior. Business rules cause things to happen, prevent things from happening, or suggest that it might be a good idea if something did or did not happen. Business rules help shape judgment, help make decisions, help evaluate, help shape behavior, and help reach conclusions.

Business rules arise from the best practices of knowledgeable business professionals. A business rule is a rule that describes, defines, guides, controls, suggests, influences or otherwise constrains some aspect of knowledge or structure within some problem domain.

Don't make the mistake of thinking that business rules are completely inflexible and that you cannot break rules. Sure, maybe there are some rules that can never be broken. Maybe there are some rules that you can break. It helps to think of breaking rules as penalties in a football game. The point is that the guidance, control, suggestions, and influence offered by business rules are a choice of business professionals. The meaning of a business rule is separate from the level of enforcement someone might apply to the rule.

A business rule states a fact about the world (declarative rule). A business rule can provide instructions (production rule). Business rules are metadata.

⁷¹ Anarchy definition, Merriam-Webster, <http://www.merriam-webster.com/dictionary/anarchy>

⁷² *Understanding that Business Rules Prevent Anarchy*, <http://xbrl.squarespace.com/journal/2016/7/15/understanding-that-business-rules-prevent-anarchy.html>

Need for Unique Handles to Identify Something

A handle can be thought of as a reference to something. In order to grab something, a computer needs to understand exactly what to grab. Suppose you were using some name to identify what to grab, if that name is not unique in the set of things you are searching to find what you want, a computer software application will be incapable of grabbing the right thing.

There are two important terms which should be understood, isomorphic and polymorphic⁷³.

- **Isomorphic** comes from the Greek term "iso" meaning "one". As used here isomorphic means that something has exactly one meaning.
- **Polymorphic** comes from the Greek term "poly" meaning more than one, many. As used here polymorphic means that something has more than one meaning.

The point here is that for a computer software application to accurately grab a piece of information that piece of information must be uniquely identifiable. The easier it is to obtain that unique identifier, or handle, the easier it is to grab the piece of information.

Artificial Intelligence

Artificial intelligence⁷⁴ is a branch of computer science. There are many good descriptions of artificial intelligence⁷⁵. Here is one good definition:

Artificial intelligence is the automation of activities that we associate with human thinking and activities such as decision making, problem solving, learning and so on.

Another more neutral term than artificial intelligence is the term machine intelligence⁷⁶ which has the same meaning. Think of these tools as narrowly focused employees with great memories that are very good at performing one specific repetitive task well, over, and over, and over. Literally, like a machine.

Those trying to make artificial intelligence work over the past 40 or so years have had limited success⁷⁷. But that is changing. People are putting the pieces together and the technologies created from AI research are now available to experiment with. If expectations are not set too high, very useful functionality for limited, narrow problems can be successfully created.

⁷³ Hypercubes/Dimensions Analysis: Summary, <http://www.xbrlsite.com/Examples/Dimensions/Index.html>

⁷⁴ Introduction to Artificial Intelligence Terminology, <http://xbrl.squarespace.com/journal/2016/7/21/introduction-to-artificial-intelligence-terminology.html>

⁷⁵ Alan Turing.net, What is Artificial Intelligence?, http://www.alanturing.net/turing_archive/pages/reference%20articles/What%20is%20AI.html

⁷⁶ Shivon Zilis and James Cham, Harvard Business Review, *The Competitive Landscape for Machine Intelligence*, <https://hbr.org/2016/11/the-competitive-landscape-for-machine-intelligence>

⁷⁷ John F. Sowa, *Why Has AI Failed? And How Can it Succeed?*, <http://www.ifsowa.com/pubs/micai.pdf>

One good example of using artificial intelligence is driverless cars. Many people get confused as to what is truly achievable and practical when it comes to driverless cars. Driverless cars are on the streets of Singapore today⁷⁸. While still in prototype mode now to work out details, by 2018 these taxis are anticipated to be commercially available. Uber is testing autonomous cars in Pittsburgh⁷⁹. Attempting to have a driverless car navigate the streets of a large city is a different problem than having a driverless truck navigate from one major city to another city on a freeway.

One of the best ways to understand the true capabilities of artificial intelligence is to try it out. Go test drive a Tesla which has driver-assist features.

Artificial intelligence programs that provide expert-level proficiency in solving problems by bringing to bear a body of knowledge about specific tasks are called knowledge based systems or expert systems. Artificial intelligence programs work using metadata.

There are two types of artificial intelligence, specialized and generalized:

- **Specialized:** An example of specialized artificial intelligence is programming a computer to play chess. The software performs one specific task. Specialized artificial intelligence is fairly easy to achieve.
- **Generalized:** An example of generalized artificial intelligence is the sort of science-fiction stuff you see in the movies. With generalized artificial intelligence, computers can perform multiple tasks or be generally intelligent. Why do you only see this in the movies? Because generalized artificial intelligence is extremely hard to make work.

Generalized artificial intelligence is 50 to 100 years in the future if you ask most experts. And general intelligence might not even be achievable. But specialized artificial intelligence is here today.

Expert Systems

Expert systems⁸⁰ is a branch of artificial intelligence. Expert systems, also called knowledge based systems or simply knowledge systems, are computer programs. The following is one definition of an expert system:

Expert systems are computer programs that are built to mimic human behavior and knowledge. Expert systems are for reconstructing the expertise and reasoning

⁷⁸ Reuters, *First driverless taxis hit the streets of Singapore*, <http://www.reuters.com/article/us-autos-selfdriving-singapore-idUSKCN1100ZG>

⁷⁹ The Atlantic, *Anybody Can Test a Driverless Car in Pennsylvania*, <http://www.theatlantic.com/technology/archive/2016/09/anybody-can-test-a-self-driving-car-in-pennsylvania/499667/>

⁸⁰ *Understanding the Components of an Expert System*, <http://xbri.squarespace.com/journal/2016/5/24/understanding-the-components-of-an-expert-system.html>

capabilities of qualified experts within some limited, narrow domain of knowledge in machine-readable form. A model of the expertise of a domain of knowledge of the best practitioners or experts is formally represented in machine-readable form and the expert system reaches conclusions or takes actions based on that information when trying to solve some problem. The computer program performs tasks that would otherwise be performed by a human expert.

Expert systems are the most commercially successful applications of artificial intelligence research⁸¹. There are currently thousands of expert systems employed world-wide in industry and government. Expert systems are driven by metadata.

Frank Puppe explains in his book *Systematic Introduction to Expert Systems*⁸² that there are three general categories of expert systems:

- **Classification or diagnosis type:** helps users of the system select from a set of given alternatives. The system tends to be instructional in nature.
- **Construction type:** helps users of the system assemble something from given primitive components.
- **Simulation type:** helps users of the system understand how some model reacts to certain inputs or create predictions based on the system.

The assembly of a financial report can be assisted by a construction-type expert system. Helping professional accountants understand what goes into that financial report can be assisted by a classification-type expert system. Creating forecasts and projections of future financial reports can be assisted by simulation-type expert systems.

Business Rules Engine

A business rules engine is a style of writing software. Anything achievable by a business rules engine is achievable by general functional programming. But, business rules engines can have an advantage in certain cases. For example, when you want to separate business logic from program logic and allow business professionals to control and maintain business rules.

A business rules engine, or rules engine, is a mechanism for executing business rules. Rules engines are optimized to figure out what rules apply to what facts and the order rules should be run and result is some consequence based on the facts and rules. A rules engine is a non-trivial application to create. The stronger the problem solving logic of the rules engine supports, the more powerful the rules engine.

⁸¹ Edward Feigenbaum et. al, *KNOWLEDGE-BASED SYSTEMS IN JAPAN*, <http://www.wtec.org/loyola/kb/execsum.htm>

⁸² Frank Puppe, *Systematic Introduction to Expert Systems, Knowledge Representations and Problem-Solving Methods*, page 11 (Note that you can read Parts I and II on Google Books here, <https://books.google.com/books?id=kKqCAAQBAJ>)

Creating an Expert System or Knowledge Based System

Creating a knowledge based system⁸³ involves the transformation of machine-readable instructions in such a way as to explain to a machine how a system works and how to make a system work the way you want that system to work.

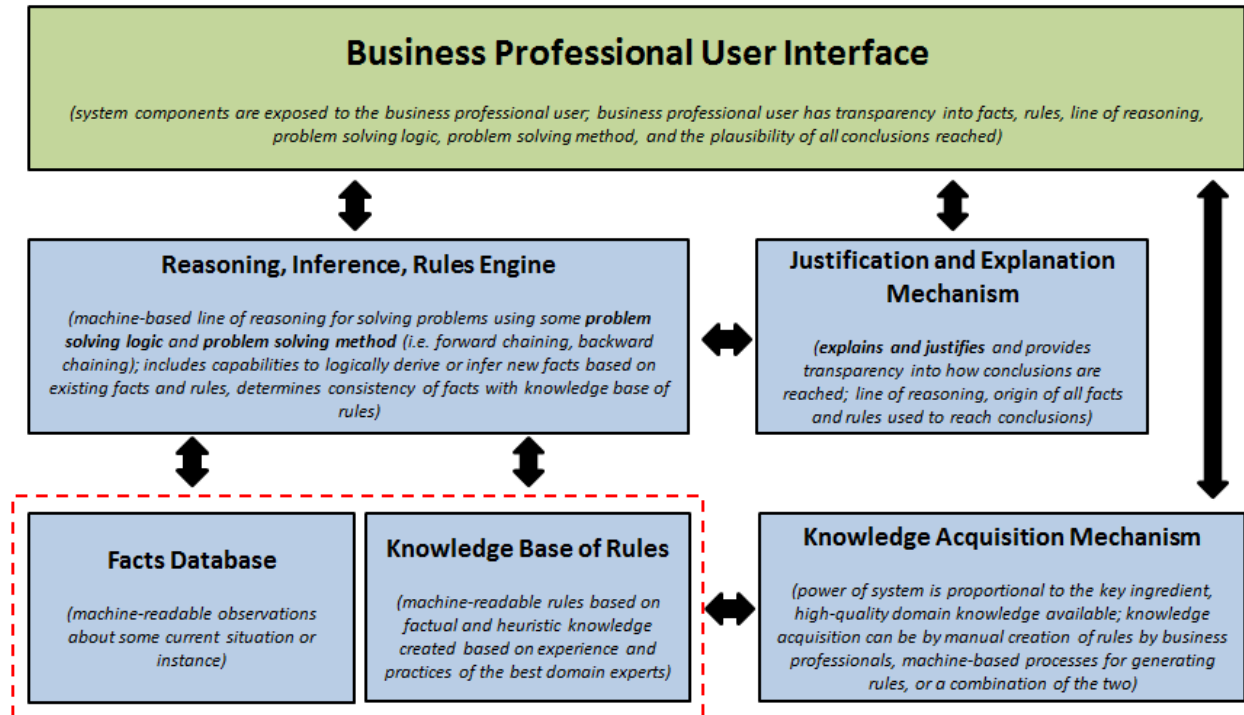
Then, brick-by-brick, much like building a house, business domain experts and software engineers can create tools that automate certain types of tasks in that process. Humans encode information, represent knowledge, and share meaning using machine-readable patterns, languages, and logic.

Components of a Knowledge Based System (Expert System)

Simply put, a **knowledge based system** is a system that draws upon the knowledge of human experts that has been represented in machine-readable form and stored in a **fact database** and **knowledge base**. The system applies **problem solving logic** using a **problem solving method** to solve problems that normally would require human effort and thought to solve. The knowledge based system supplies an **explanation and justification mechanism** to help system users to understand the **line of reasoning** used and support **conclusions reached** by the knowledge based system and presents that information to the user of the system.

Nothing is a “black box”, rather every aspect of every decision made by a system is completely understandable to the business professional using the system. Below is a graphic of the components of a knowledge based system:

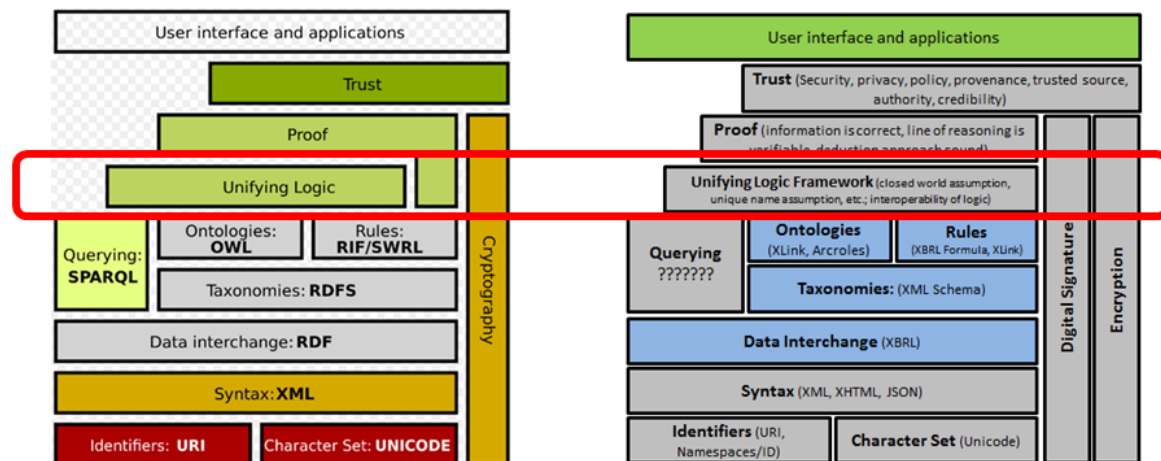
⁸³ Another term used to describe such systems is reasoning system.



Multiple Technology Stacks

As was stated earlier; arbitrary preferences, fads, trends, misinformation, and many other things influence how information technology professionals solve the same problem. As such, there will always be multiple technology stacks as opposed to every information technology professional using exactly the same approach to solving what amounts to be exactly the same problem.

Consider the comparison of these two technology architecture stacks⁸⁴: Semantic Web Stack and XBRL Stack:



⁸⁴ Unifying Logic Framework for Business, <http://xbrl.squarespace.com/journal/2017/11/26/unifying-logic-framework-for-business.html>

Each of the technical architecture stacks has a problem solving logic. The Semantic Web Stack on the left calls it a “Unifying Logic” and the XBRL Stack calls it a “Unifying Logic Framework”⁸⁵.

The problem solving logic for the Semantic Web Stack is defined by the technical syntax offered by RDFS, OWL, RIF/SWIRL, SHACL. For the Semantic Web Stack, those technical syntaxes provide the boundaries for the problem solving logic. For the XBRL Stack, there really are no real boundaries outlined at all for the problem solving logic. Even if someone could provide the specifics of the problem solving power offered by RDFS, OWL, and RIF/SWIRL, SHACL; that list undoubtedly would not be understandable to professional accountants or other business professionals. But if business professionals are supposed to control and maintain business rules, then those business logic boundaries would best be clear and understandable. The bottom line here is that the boundaries of a problem solving logic should be understandable.

Representing business domain knowledge

Skilled craftsmen using the right tools can get the dumb beasts to perform magic. But it is not really magic that is occurring; it just seems like magic. What is really going on is shared realities are being created in order to leverage to capabilities of these machines in order to make things better, faster, and cheaper. The goal is to increase productivity.

Shared View of Reality to Achieve a Specific Purpose

In his book⁸⁶ *Data and Reality*, William Kent provides an excellent summary that discusses the realities of sharing information. In Chapter 9: *Philosophy* in the Third Edition and Chapter 12: *Philosophy* in the first edition (which is available online) he paints a picture of why you want to go through the trouble of sharing information using machine-based processes and the realities of what that takes. This is what William Kent points out which I have paraphrased as it relates to financial reporting:

To create a shared reality to achieve a specific purpose: To arrive at a shared common enough view of "true and fair representation of financial information" such that most of our working purposes, so that reality does appear to be objective and stable so that you can query information reliably, predictably, repeatedly, safely.

Meaningful information exchange that is reliable, repeatable, predictable, safe, cost effective, easy to use, robust, scalable, secure when necessary, auditable (track provenance) when necessary.

⁸⁵ *Unifying Logic Framework for Business*, <http://xbrl.squarespace.com/journal/2017/11/26/unifying-logic-framework-for-business.html>

⁸⁶ William Kent, *Data and Reality*, Technics Publications, (See this resource which has CHAPTER 12: Philosophy from the first version of this book, <http://www.bkent.net/Doc/darxrp.htm>)

Prudence dictates that using the information contained in a digital financial report should not be a guessing game. Safe, reliable, repeatable, predictable, reuse of reported financial information using automated machine-based processes is obviously preferable to a guessing game.

The effective meaningful exchange of information is created by skilled craftsmen that know their craft well. The craftsmen balance the system, bringing the system into equilibrium to achieve some specific purpose. Creating this shared view of reality which allows this specific purpose to be achieved has benefit to the financial reporting supply chain.

That purpose should be clearly defined so that everyone understands the objective and exactly what the system can, and cannot, deliver.

Principles help you think about something thoroughly and consistently. Overcoming disagreements between stakeholders and even within groups of stakeholders is important. Agreement between stakeholder groups and within stakeholder groups contributes to harmony. Lack of agreement contributes to dissonance. Principles help in the communications process.

I would argue that a first step, if not the first step, of arriving at harmony is outlining the interests, perceptions, positions, and risks of each constituency/stakeholder group.

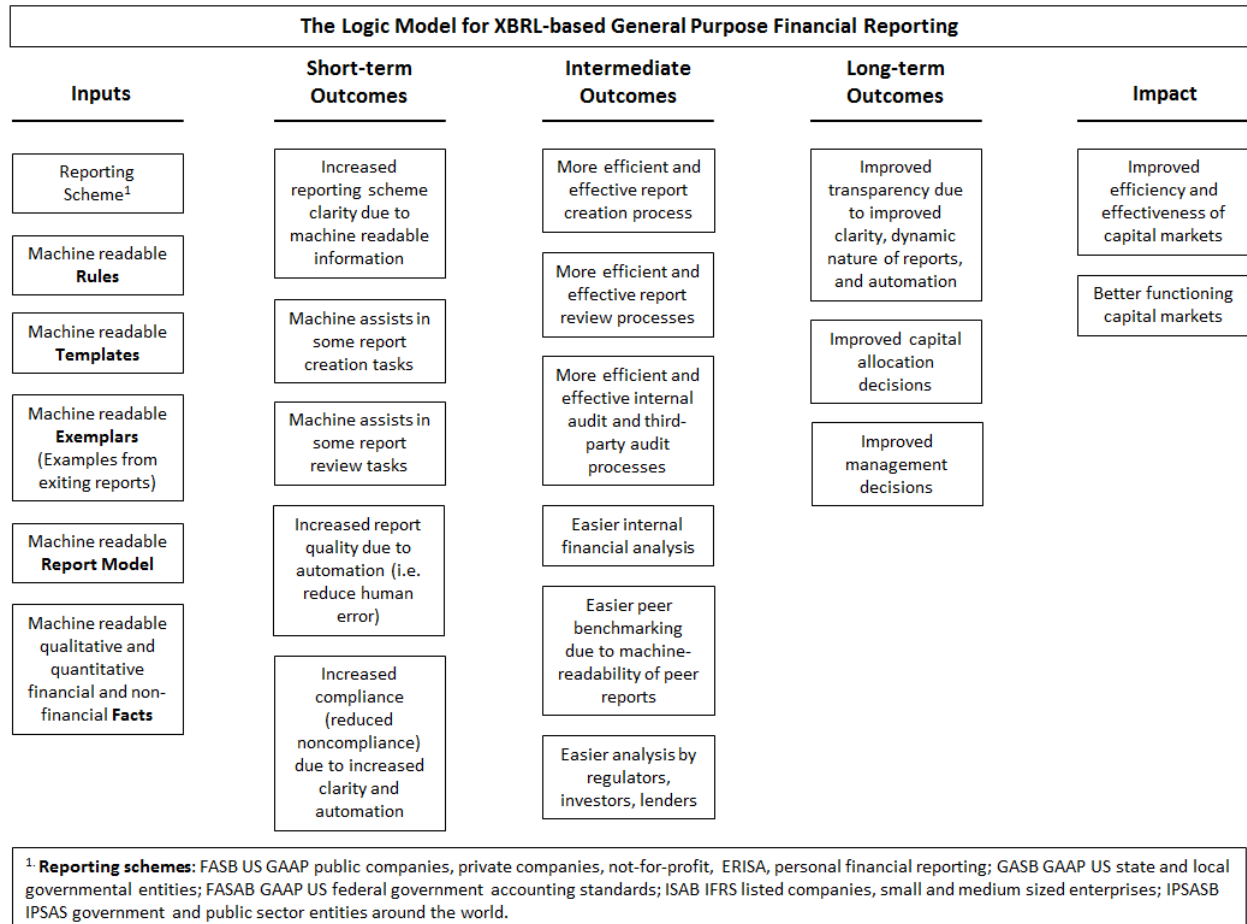
A "stakeholder" is anyone that has a vested interest. Another term for stakeholder is "constituent". A "constituent" is a component part of something.

Foundational to arriving at harmony is having a common conceptual framework including a set of consistent principles or assumptions or world view for thinking about the system. For example, accounting and financial reporting have such a conceptual framework including principles/assumptions such as "materiality" and "going concern" and "conservatism".

This "framework for agreeing" helps the communications process which increases harmony and decreases dissonance. This is about bringing the system into balance, consciously creating the appropriate equilibrium/balance.

Case for XBRL-based General Purpose Financial Reporting on One Slide

The following graphic summarizes the case for and benefits of XBRL-based general purpose financial reporting in one slide:



What would it take to make XBRL-based general purpose financial reporting work effectively? What if all the short-term outcomes, intermediate outcomes, long-term outcomes could be realized and the impact of more efficient and effective and better functioning capital markets could result? What if the process of creating, reviewing, auditing, and analyzing a financial report could actually be made better, faster, and cheaper?

There is no need to speculate about this possibility. If you see software that is better, faster, and cheaper you will know it. To create this software you need to harness the capabilities of computers.

Business Rules Should Be Controlled and Maintained by Business Professionals

Business professionals create and maintain business rules. As Article 9 of the *Business Rules Manifesto*⁸⁷ states, business rules are of, by, and for business people; not information technology people.

- 9.1. Rules should arise from knowledgeable business people.
- 9.2. Business people should have tools available to help them formulate, validate, and manage rules.
- 9.3. Business people should have tools available to help them verify business rules against each other for consistency.

Rather than creating tools that only information technology professionals can use because the tools are so complicated; business professionals need to demand software tools that properly expose functionality that exposes business rules to business users such that business users are working with business domain knowledge, not technical details that should have been buried deeply within the software applications.

Business professionals need to understand the *Law of Conservation of Complexity*⁸⁸ which states that complexity can never be removed from a system, but complexity can be moved. The *Law of Conservation of Complexity* states: "Every application has an inherent amount of irreducible complexity. The only question is: Who will have to deal with it - the user, the application developer, or the platform developer?"

Categories of Business Rules

At their essence, business rules articulate information about something or about the relationship between one thing and some other thing. Some examples that can help you better understand exactly what business rules are:

- **Assertions:** For example asserting that the balance sheet balances or "Assets = Liabilities + Equity".
- **Computations:** For example, calculating things, such as "Total Property, Plant and Equipment = Land + Buildings + Fixtures + IT Equipment + Other Property, Plant, and Equipment".
- **Constraints:** For example, specific behavioral constraints that control when it is appropriate to create, update, or remove information.

⁸⁷ *Business Rules Manifesto*, <http://www.businessrulesgroup.org/brmanifesto.htm>

⁸⁸ Understanding the Law of Conservation of Complexity, <http://xbrl.squarespace.com/journal/2015/5/24/understanding-the-law-of-conservation-of-complexity.html>

- **Continuity cross-checks:** For example, if a fact is used that fact does not conflict with or contradict other facts.
- **Process-oriented rules:** For example, the disclosure checklist commonly used to create a financial statement which might have a rule, "If the line item Property, Plant, and Equipment exists on the balance sheet, then a Property, Plant and Equipment policies and disclosures must exist."
- **Regulations:** Another type of rule is a regulation which must be complied with, such as "The following is the set of ten things that must be reported if you have Property, Plant and Equipment on your balance sheet: depreciation method by class, useful life by class, amount under capital leases by class ..." and so on. Many people refer to these as reportability rules or statutory and regulatory compliance requirements.
- **Instructions or documentation:** Rules can document relations or provide instructions, such as "Cash flow types must be either operating, financing, or investing."
- **Relations:** How things can be related, such as whole-part relations. For example, how the business segments of an economic entity are related.

Scaling Business Rules

Managing business rules becomes more complex as the number of rules increase. Scaling business rules is important. Using a decision model based approach can help manage large sets of business rules. The article, *How DMN Allows Business Rules to Scale*⁸⁹ points out four primary problems that you run into:

- **The 'Rush to Detail':** Business rule development encourages policy makers to focus on rule implementation prematurely, before they have considered the broader goals and structure of their business decisions and to what extent they will be automated. This approach is like starting to build a house by laying bricks, rather than drawing plans and establishing foundations.
- **Poor Dependency Management:** A growing and poorly understood set of inter-dependencies between rules causing changes to have unintended consequences—making the rule set brittle and reducing its agility.
- **Insufficient Transparency:** The bewildering size of a rule set, use of technical (rather than business) terms and style for expressing rules and a poor connection between rules and their business context (their rationale and place in the business process)—making the meaning and motivation of rules more obscure.

⁸⁹ Jan Purchase, *How DMN Allows Business Rules to Scale*, November 19, 2015, <http://blog.luxmagi.com/2015/11/how-dmn-allows-business-rules-to-scale/>

- **Lack of Growth Management:** Poor discipline about the scope, quality and placement of rules that are added to the rule set—making it hard to find rules and leading to ‘stale’ rules and duplicates.

Separating Business Rules from Code

Historically, business logic and application logic have been intermingled in software algorithms. Whenever business logic changed, it took a programmer to make the change in business logic in software algorithms. To do that, business professionals had to explain the new business logic to programmers then programmers would need to make the change in algorithms.

But that is changing. Business logic and application logic should be separated. Business professionals should be able to control business logic and change aspects of how software works by changing the business logic as business needs dictate.

Separating business logic from application logic is achieved by representing business rules that control the business logic of software in machine-readable form that is understandable by and under the control of business professionals. Then, a different type of software can be created and used; a business rules engine.

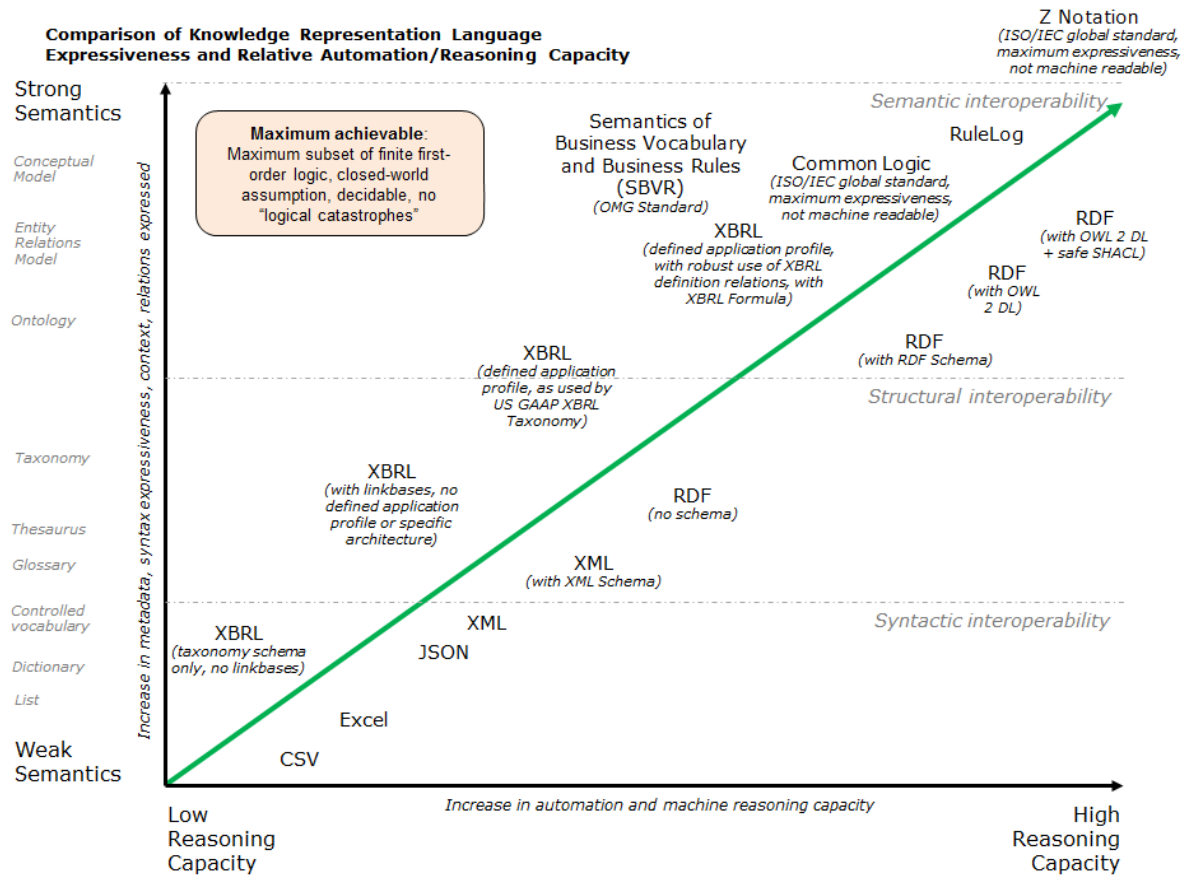
Problem Solving Method

There are basically two primary problem solving methods: forward chaining and backward chaining. Each method has pros and cons. Forward chaining can do everything that backward chaining can do; but backward chaining cannot do everything that forward chaining can do. It is best to use both approaches, applying each when circumstances dictate⁹⁰.

Problem Solving Logic

Problem solving logic can best be described by two things: (a) the expressive power of the logic and (b) the safety of the logic to avoid catastrophic failure. The power of a problem solving logic is directly related by an ability to express rules. Different technical syntax support different problem solving logic. The following graphic compares and contrasts the relative expressive power of different syntax that can be used to express rules.

⁹⁰ *Differentiating Forward and Backward Chaining*, <http://xbrl.squarespace.com/journal/2016/5/27/differentiating-forward-and-backward-chaining.html>



Inspired by similar comparisons from *An Intrepid Guide to Ontologies* <http://www.mkbergman.com/date/2007/05/16/> and *Semantics Overview* <http://prezi.com/prwxi8po3ln/semantics-overview/>

Logic Representation Approaches

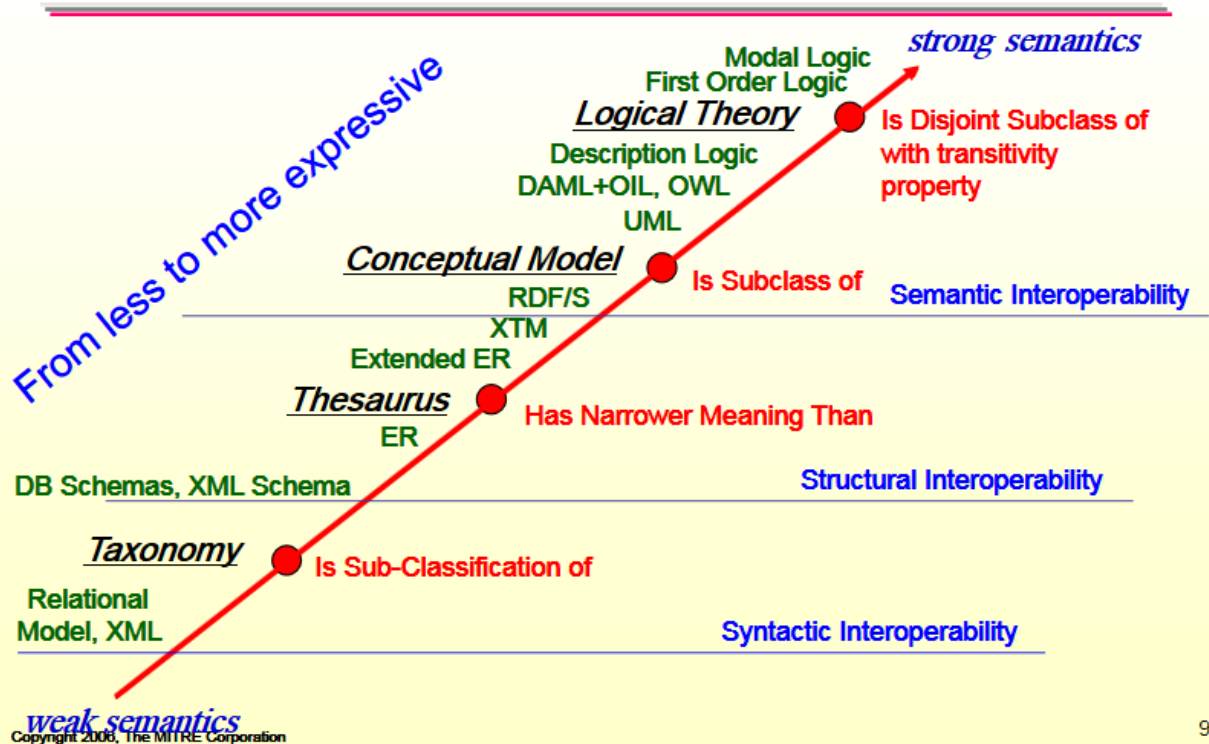
There are many different approaches that can be used to represent logic in machine readable form. Each approach has pros and cons. I have mentioned taxonomies and ontologies. There are entity relationship models, conceptual models, you can use the Unified Modeling Language (UML) or a spinoff of UML called SysML⁹¹. The diagram above points out approaches such as RuleLog, Common Logic, Semantics of Business Vocabulary and Business Rules (SBVR), Z Notation. The one thing that each of these approaches have in common is that the typical business professional cannot understand any of these approaches.

Dr. Leo Obrst of Mitre published *The Ontology Spectrum and Semantic Models*⁹² which provides the following slide:

⁹¹ SysML, <https://sysml.org/>

⁹² Dr. Leo Obrst, *The Ontology Spectrum and Semantic Models*, slide 9, <https://slideplayer.com/slide/697642/>

Ontology Spectrum: One View



The slide in general and the presentation in particular points out tools that are available for representing logic, or semantics, in machine readable form which can then be used by computer processes. Not all approaches are the same. Note the term “Logical Theory” and how high that approach is in terms of strong semantics. I will come back to that in a moment, but first three important ideas.

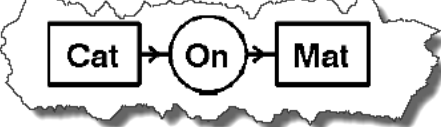
Technical Syntax is Not Really Relevant to Business Professionals

Arbitrary preferences, fads, trends, misinformation, and many other things influence how information technology professionals solve the same problem. How a problem is solved by information technology professionals is less important to business professionals. What is important is for business professionals to properly define a problem and having the maximum capabilities to solve the problem, maximum safety of the system, and maximize the usability of the system by business professionals.

- **Syntax** is how you say something
- **Semantics** is the meaning behind what you said, the business logic

How to Say “A cat is on a mat.”

John Sowa provides a simple example that makes a profound point. How to you represent the notion that “a cat is on a mat” in various technical syntax forms⁹³. Here are the examples he provides:

	$\Sigma_x \Sigma_y \text{Cat}_x \bullet \text{Mat}_y \bullet \text{On}_{x,y}$
$\exists x \exists y \text{Cat}(x) \wedge \text{Mat}(y) \wedge \text{On}(x,y)$	<pre>SELECT FIRST.ID, SECOND.ID FROM OBJECTS FIRST, OBJECTS SECOND, SUPPORTS WHERE FIRST.TYPE = "Cat" AND SECOND.TYPE = "Mat" AND SUPPORTS.SUPPORTER = SECOND.ID AND SUPPORTS.SUPPORTEE = FIRST.ID</pre>
$(\text{exists } ((x \text{ Cat}) (y \text{ Mat})) (\text{On } x \ y))$	$[\text{Cat } *x] [\text{Mat } *y] (\text{On } ?x \ ?y)$
	$\text{A cat is on a mat.}$

Which of these syntax do you find the easiest to read? Which syntax conveys the most meaning to you? Which of these technical syntax would be easiest for a business professional to work with? How hard is it to convert one syntax to another syntax if the logic is the same? One approach is to use logic symbols⁹⁴ to represent words. Another approach is to use a controlled natural language to explain logic.

Differences between Ontologies, Rules, and Schemas

Ontologies and business rules are overlapping rather than disjointed⁹⁵. Ontologies, rules, and schemas are all approaches to representing knowledge and each approach has strengths and

⁹³ John F. Sowa, *Common Logic: A Framework for a Family Of Logic-Based Languages*, page 2-3, <http://www.ifsowa.com/ikl/SowaST08.pdf>

⁹⁴ Wikipedia, *Logic Symbols*, https://en.wikipedia.org/wiki/List_of_logic_symbols

⁹⁵ Benjamin Grosz, *Survey of Knowledge Representations for Rules and Ontologies*, page 4, <http://coherentknowledge.com/wp-content/uploads/2013/05/Ontolog-Forum-talk-OF-surveyKR-20131024-BNG.pdf>

weaknesses⁹⁶. The primary challenge relating to figuring out the best approach to use to represent knowledge, as pointed out by John Sowa in his paper *Fads and Fallacies about Logic*⁹⁷, is not technical at all. Most issues relate to an even more formidable challenge related to fads, trends, arbitrary preferences, politics, fallacies, misinformation, and alternative completing standards.

Approaches to representing knowledge can be placed into three broad groups: *ontologies*, *rules*, and *schemas*. The purpose of each of these is to represent some problem domain or conceptual model of some sort. The following is a summary of the essence of each of these three approaches to representing knowledge:

- **Ontology:** An ontology is generally a subset of knowledge that is definitional in nature and focuses on defining terminology and categories/subcategories/properties of terms and the relations between the categories/subcategories/properties. An ontology is a collection of taxonomies. Ontologies are axiom-based.
- **Rules:** Rules are assertions or statements which have an IF...THEN format that have logical implication. The set of rules can be definitional in nature and therefore in essence ontological. A rule (or assertion, statement) can be TRUE or it can be FALSE. Rules are model-based.
- **Schema:** A schema is an outline, diagram, or model that tend to describe structure. To examples of schemas are a database schema which describes the structure of data in a database and XML Schema which is used to describe the structure of an XML document.

Too many people are in either the “ontology” camp or the “rules” camp or the “schema” camp to the exclusion of all other alternatives which tends to create silos and each camp acting dogmatic. This is not helpful to business professionals trying to solve problems practically.

Business professionals should focus on logic and the problem solving logic provided by a system rather than the implementation details. If rich enough, an ontology, a set of rules, or a schema can be used to represent a conceptual model.

Logical Theory

A **theory** is a formal statement of rules about some subject that describes and otherwise explains the nature of that subject. A theory describes some aspect of the world and tries to describe the principles by which that aspect of the world operates. A theory can be right or wrong, but a theory is characterized by its intent: the discovery of essence. A theory does not

⁹⁶ Silvie Spreeuwenberg, *The Semantic Web and the Business Rules Approach ~ Differences and Consequences*, <https://www.brcommunity.com/articles.php?id=b257>

⁹⁷ John F. Sowa, *Fads, Misinformation, Trends, Politics, Arbitrary Preferences, and Standards*, <http://www.ifsowa.com/pubs/fflogic.pdf>

simplify. A theory describes absolutes. A successful theory can become a fact. A theory is a tool for contemplating something with an intent to gain insight or understanding.

Logic a set of principles that form a framework for correct reasoning. Logic is a process of deducing information correctly. Logic is about the correct methods that can be used to prove a statement is true or false. Logic tells us exactly what is meant. Logic allows systems to be proven.

In logic, a **statement** is a sentence that is either true or false. You can think of statements as pieces of information that are either correct or incorrect. And therefore, statements are pieces of information that you apply logic to in order to derive other pieces of information which are also statements. Statements are basically rules.

A **logical theory** is a set of logical statements that formally describes some subject or system. **Axioms**⁹⁸ are statements that describe self-evident logical principles that no one would argue with. **Theorems**⁹⁹ are logical deductions which can be proven by constructing a chain of reasoning by applying axioms and the rules of logic in the form of IF...THEN statements.

A **rule**, or business rule or assertion, is a true statement with respect to some model of the real world that could possibly exist given some logical theory. You cannot create rules that are true in worlds that can never exist. A rule can be a mathematical expression. A rule is a type of logical statement.

The ***Financial Report Semantics and Dynamics Theory***¹⁰⁰ is a logical theory that explains how the mechanical aspects of a financial report work. The ***Logical Theory Describing a Business Report***¹⁰¹ explains how a general business report works. Both are based on the same ideas and tend to be significantly more understandable than other approaches for explaining the underlying logic of a system.

Using Conceptual Models

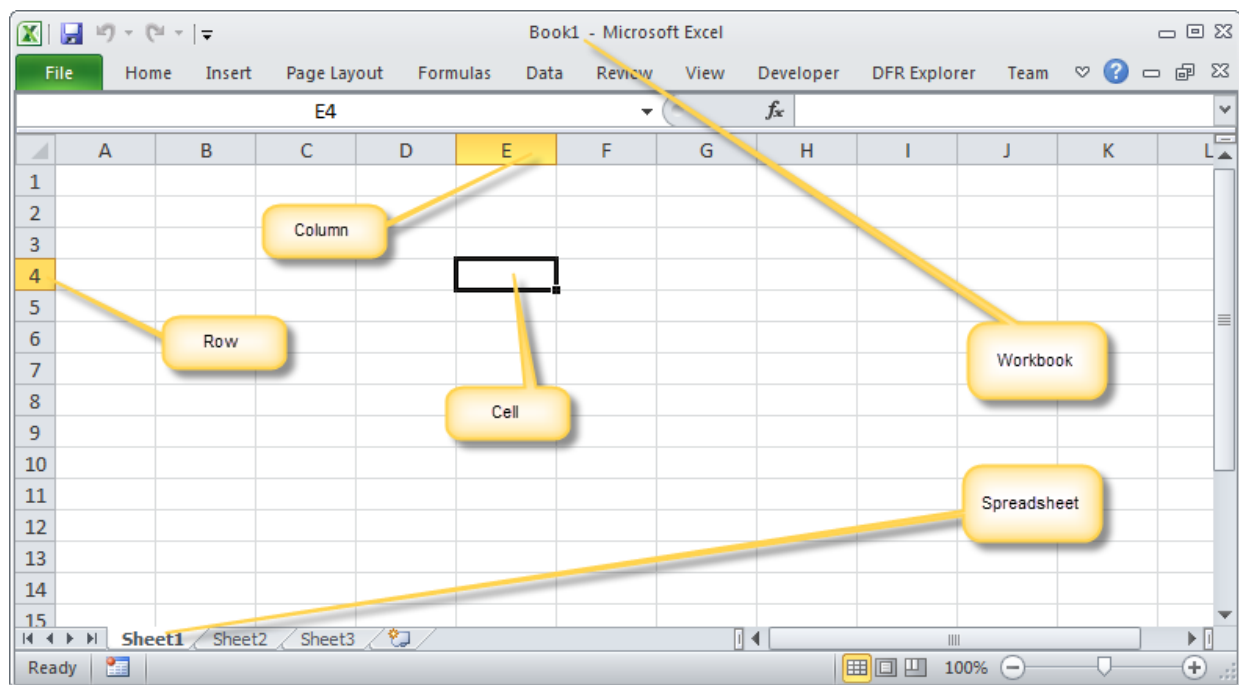
Business professionals work with conceptual models every day. For example, the workbooks, spreadsheets, rows, columns, and cells of an electronic spreadsheet are a conceptual model. The ease and simplicity of an electronic spreadsheet allows the average business professional to make use of this helpful tool.

⁹⁸ Wikipedia, *Axiom*, <https://en.wikipedia.org/wiki/Axiom>

⁹⁹ Wikipedia, *Theorem*, <https://en.wikipedia.org/wiki/Theorem>

¹⁰⁰ Charles Hoffman, CPA and Rene van Egmond, *Financial Report Semantics and Dynamics Theory*, <http://xbrlsite.azurewebsites.net/2016/Library/Theory-2017-06-26.pdf>

¹⁰¹ Charles Hoffman, CPA and Rene van Egmond, *Logical Theory Describing a Business Report*, <http://xbrlsite.azurewebsites.net/2019/Library/LogicalTheoryDescribingBusinessReport.pdf>



Differentiating a Notion/Idea/Phenomenon, a Name, and a Preferred label

It is important to understand and properly differentiate between the following three things:

- **Notion, idea, phenomenon:** something that exists in reality that needs to be represented
- **Name:** helps computers uniquely identify some notion/idea/phenomenon that is a representation of reality within some machine-readable conceptual model
- **Preferred label:** alternative ways used to refer to name

Confusing these three things can cause problems when trying to create a conceptual model. Two things that are genuinely different should have two different names. However, if one thing is given two names when the one thing really is two different preferred labels problems can occur.

Power of Agreement

It is only through deliberate, methodical, rigorous and conscious collaboration, cooperation and coordination by the participants of the financial reporting supply chain that XBRL-based digital financial reporting will work safely, reliably, predictably, repeatedly, effectively, and efficiently. This objective will not be achieved by accident.

Consider the definitions of arbitrary and standard:

- **Arbitrary:** based on random choice or personal whim, rather than any reason or system; depending on individual discretion (as of a judge) and not fixed by law
- **Standard:** used or accepted as normal; something established by authority, custom, convention, law, regulation, or general consent as a model or example

US GAAP contains many, many standard terms. For example, Equity, Assets, Liabilities, etc. The US GAAP XBRL Taxonomy names these terms, providing a standard. A common obstacle to creating a working dictionary of concepts and relations between those concepts is disagreement as to those definition and relations. Agreement by all stakeholders through deliberate, methodical, rigorous and conscious collaboration, cooperation, and coordination can help overcome this obstacle.

Differentiating the Important from the Unimportant

The following terms help one understand the difference between an important nuance and an unimportant negligible difference.

- **Nuance:** a subtle but important difference in or shade of meaning, expression, or sound; a subtle but important distinction or variation
- **Subtle:** so delicate or precise as to be difficult to analyze or describe but important; hard to notice or see but important; not obvious but important
- **Negligible:** so small or unimportant as to be not worth considering; insignificant; so small or unimportant or of so little consequence as to warrant little or no attention

Business professionals can best differentiate important nuances from unimportant negligible differences. They do not do it perfectly and the only real way to make sure things are right is testing and experimentation.

Conceptual models, ontologies, and theories are about getting the salient aspects of a problem domain right. One needs to take a pragmatic view of the world because it is impossible to describe every single aspect of the real world. Such frameworks only need to represent the important things and serve as a “wireframe” or a “substrate” of reality. Getting bogged down in unimportant, insignificant, or inconsequential details at best serves no purpose, at worst can cause unnecessary complexity.

Difference between Simplistic and Simple

Anyone can create something that is sophisticated and complex. It is much harder to create something that is sophisticated and simple. Simple is not the same thing as simplistic. "Simple" is not about doing simple things. Simple is the ultimate sophistication. Simple is elegant.

- **Simplicity:** Simplicity is “dumbing down” a problem to make the problem easier to solve.
- **Simple:** Simple is about beating down complexity in order to make something simple and elegant; to make sophisticated things simple to use rather than complex to use.

Creating something that is simple takes conscious effort and is hard work.

Difference between a Requirement and a Policy

Sometimes things are required, other times things are a choice. Yet in other times setting some policy eliminates certain options which could have been previously considered.

- **Policy:** a course or principle of action adopted or proposed by a government, party, business, or individual; definite course or method of action selected from among alternatives and in light of given conditions to guide and determine present and future decisions
- **Requirement:** a thing that is needed or wanted; something that is essential or that must be done
- **Choice:** an act of selecting or making a decision when faced with two or more possibilities; the act of choosing; the act of picking or deciding between two or more possibilities
- **Option:** a thing that is or may be chosen; the opportunity or ability to choose something or to choose between two or more things

Any time a business professional is presented with an alternative complexity increases because the business professional then must choose between the available alternatives. As the number of choices increases, complexity increases. Choices must be managed. Flexibility when it is not necessary is not a feature, it is a bug.

Best Practices

A best practice is a method or technique that has been generally accepted as superior to any other known alternatives because it produces results that are superior to those achieved by other means or because it has become a standard way of doing things.

High-fidelity, High-resolution, High-Quality

As stated, a general purpose financial report provides high-fidelity, high-resolution information that is of very high-quality. Consider this scenario:

Two public companies, A and B, each have knowledge about their financial position and financial performance. They must communicate their knowledge to an investor who is making investment decisions which will make use of the combined information so as to

draw some conclusions. All three parties are using a common set of basic logical principles (facts known to be true, deductive reasoning, inductive reasoning, etc.) and common financial reporting standards (i.e. US GAAP, IFRS, etc.), so they should be able to communicate this information fully, so that any inferences which, say, the investor draws from public company A's input should also be derivable by public company A using basic logical principles and common financial reporting standards, and vice versa; and similarly for the investor and public company B.

Let's be clear about the terms we are using and the need for low to zero tolerance for error. Specifically, let's be clear about the following definitions:

- **Reliability** is about getting consistent results each time an activity is repeated.
- **Accuracy** is about identifying the correct target. Accuracy relates to correctness in all details; conformity or correspondence to fact or given quality, condition; deviating within acceptable limits from a standard. Accuracy means with no loss of resolution or fidelity of what the sender wishes to communicate and no introduction of false knowledge or misinterpretation of communicated information.
- **Precision** is the closeness of repeated measurements to one another. Precision involves choosing the right equipment and using that equipment properly. Precise readings are not necessarily accurate. A faulty piece of equipment or incorrectly used equipment may give precise readings (all repeated values are close together) but inaccurate (not correct) results.
- **Fidelity** relates to the exactness or loyal adherence facts and details with which something is copied or reproduced. Fidelity relates to the faithful representation of the facts and circumstances represented within a financial report properly reflect, without distortion, reality. High fidelity is when the reproduction (a financial report) with little distortion, provides a result very similar to the original (reality of economic entity and environment in which economic entity operates).
- **Integrity** is the quality or condition of being whole or undivided; completeness, entireness, unbroken state, uncorrupt. Integrity means that not only is each piece of a financial report correct but all the pieces of the financial report fit together correctly, all things considered.
- **Resolution** relates to the amount of detail that you can see. The greater the resolution, the greater the clarity.
- **Completeness** relates to having all necessary or normal parts, components, elements, or steps; entire.
- **Correctness** relates to freedom from error; in accordance with fact or truth; right, proper. Consistency relates to being compatible or in agreement with itself or with

some group; coherent, uniform, steady. Holding true in a group, compatible, not contradictory.

Differentiating between Objective and Subjective

There is a difference between something that is objective and something that is subjective.

- **Objective:** not influenced by personal feelings or opinions in considering and representing facts; based on facts rather than feelings or opinions; not influenced by feelings; facts are objective.
- **Subjective:** based on or influenced by personal feelings, tastes, preferences, or opinions; based on feelings or opinions rather than facts; relating to the way a person experiences things in his or her own mind; opinions are subjective.
- **Judgment:** the ability to make considered decisions or come to sensible conclusions; an opinion or decision that is based on careful thought; judgment is subjective.

Remember, computers are machines. Computers have no intelligence until they are instructed by humans. Computers only appear smart when humans create standards and agree to do things in a similar manner in order to achieve some higher purpose. It is easy to agree on things that tend to be objective. It is harder to agree where there is subjectivity. It is extremely difficult to impossible to get a machine to exercise judgment. A machine such as a computer can only mimic what humans tell the machine to do via machine-readable information.

Difference between Explicit and Implicit

In the process of agreeing, it is important to understand the difference between what is important and what is unimportant in that process of agreeing. It is likewise important to understand the difference between telling a machine something and requiring the machine to figure something out:

- **Explicit:** Stated clearly and in detail, leaving no room for confusion or doubt; very clear and complete; leaving no doubt about the meaning.
- **Implicit:** Implied though not plainly expressed; understood though not clearly or directly stated.
- **Ambiguous:** Open to more than one interpretation; having a double meaning; able to be understood in more than one way; having more than one possible meaning; not expressed or understood clearly.
- **Derive or Impute:** Assign (a value) to something by inference from the value of the products or processes to which it contributes; to deduce a conclusion about some fact using some other fact or facts and logical reasoning.

Machines do well with information which is explicitly provided. When information is not explicitly provided, software developers either make a choice or have to figure out ways to allow a business professional making use of the software to make a choice. Every time a software developer or business professional has to make an interpretation because something is ambiguous, there is the possibility that some unexpected or incorrect interpretation can be made. Not being explicit causes confusion and turns using ambiguous information into a guessing game.

Multidimensional Logical Model

Logical models help communication and provide a framework for understanding. The multidimensional logical model is a model for understanding information¹⁰². Every professional accountant works with multidimensional information every day but don't generally realize it.

Just like an electronic spreadsheet has a logical model (workbook, spreadsheet, row, column, cell); a digital financial report has a logical model. The logical model of a digital financial report follows the multidimensional logical model. Here are the high-level pieces of a digital financial report:

- **Fact:** A fact defines a single, observable, reportable piece of information contained within a financial report, or fact value, contextualized for unambiguous interpretation or analysis by one or more distinguishing characteristics. Facts can be numbers, text, or prose.
- **Characteristic:** A characteristic describes a fact (a characteristic is a property of a fact). A characteristic provides information necessary to describe a fact and distinguish one fact from another fact. A fact may have one or many distinguishing characteristics.
- **Fact table:** A fact table is a set of facts which go together for some specific reason. All the facts in a fact table share the same characteristics.
- **Relation:** A relation is how one thing in a business report is or can be related to some other thing in a business report. These relations are often called business rules. There are three primary types of relations (others can exist):
 - **Whole-part:** something composed exactly of their parts and nothing else; the sum of the parts is equal to the whole (roll up).
 - **Is-a:** descriptive and differentiates one type or class of thing from some different type or class of thing; but the things do not add up to a whole.
 - **Computational business rule:** Other types of computational business rules can exist such as "Beginning balance + changes = Ending Balance" (roll forward) or "Net income (loss) / Weighted average shares = Earnings per share".

¹⁰² Introduction to the Multidimensional Model for Professional Accountants, <http://xbrl.squarespace.com/journal/2016/3/18/introduction-to-the-multidimensional-model-for-professional.html>

- **Grain:** Grain is the level of depth of information or granularity. The lowest level of granularity is the actual transaction, event, circumstance, or other phenomenon represented in a financial report. The highest level might be a line item on a primary financial statement such as a balance sheet.

Information robots

During the 1900s many routine, mundane *manual tasks* were automated using machines including sophisticated robots. Now, many routine, mundane *cognitive tasks* will be very likely be automated.

Process Robotics is Disrupting Accounting and Finance

In the video, *Finance in a Digital World and the Impact on CFOs*¹⁰³, John Steel who leads the finance transformation practice of Deloitte makes the statement,

"Five years from now there is either no CFO or the CFO is playing a different role."

John's view is consistent with what the AICPA and Journal of Accountancy are saying which is that technology is poised to change the accounting profession¹⁰⁴.

In the video, John goes on to say, "Digital is having a tremendous impact and it's quite disruptive." In the video he goes over trends that are occurring. One of those trends is process robotics. John uses the term "lights out finance" meaning a finance that is completely automated. Now, we may never get to where the lights are completely out, but digital will involve automation of many existing manual processes.

A key word here is "disruptive". As pointed out by *The Innovators Dilemma*¹⁰⁵, there are two types of innovation: sustaining and disruptive. *Sustaining innovation* meets customer's current needs, making incremental improvements in quality and efficiency of current processes. *Disruptive innovation* is about meeting future needs of customers.

Process robotics is about automating accounting, clerical, administrative, and other such tasks using software robots. Artificial intelligence technology drives these software robots. It really is a lot like how physical robots were employed to replace humans in manufacturing processes such as the process of building cars. Software robots cannot automate all tasks but certainly

¹⁰³ SAP, *Finance in a Digital World and the Impact on CFOs*, <http://events.sap.com/sapandasug/en/session/32172>

¹⁰⁴ AICPA News Update: *Technologies are poised to reshape the accounting landscape*, <http://xbrl.squarespace.com/journal/2017/7/7/aicpa-news-update-technologies-are-poised-to-reshape-the-acc.html>

¹⁰⁵ YouTube, *The Innovator's Dilemma*, <https://www.youtube.com/watch?v=vUAtIQDllo8>

there are tasks which can be automated. As an example, Blackline offers an accounting process automation product¹⁰⁶ and FlexRule offers a general process robotics platform¹⁰⁷.

We don't want to overstate what innovations such as process robotics will provide. But we likewise don't want to understate the impact either. This is not about computers taking over the world. The way it will work is that you will have humans augmented by machine capabilities, much like an electronic calculator enabling a human to do math quicker, will empower professional accountants and others who know how to leverage those machines.

Underlying the process robotics will be intelligent software agents that interact with each other and with humans to get work done.

Understanding Artificial Intelligence and Intelligent Software Agents

Artificial intelligence is the automation of activities that we associate with human thinking and activities such as decision making, problem solving, learning and so on. An intelligent software agent¹⁰⁸ is software that assists people and acts on their behalf. Intelligent agents work by allowing people to:

- delegate work that they could have done to the agent software,
- perform repetitive tasks,
- remember things you forgot,
- intelligently find, filter and summarize complex information,
- customize information to your preferences,
- learn from you and even make recommendations to you.

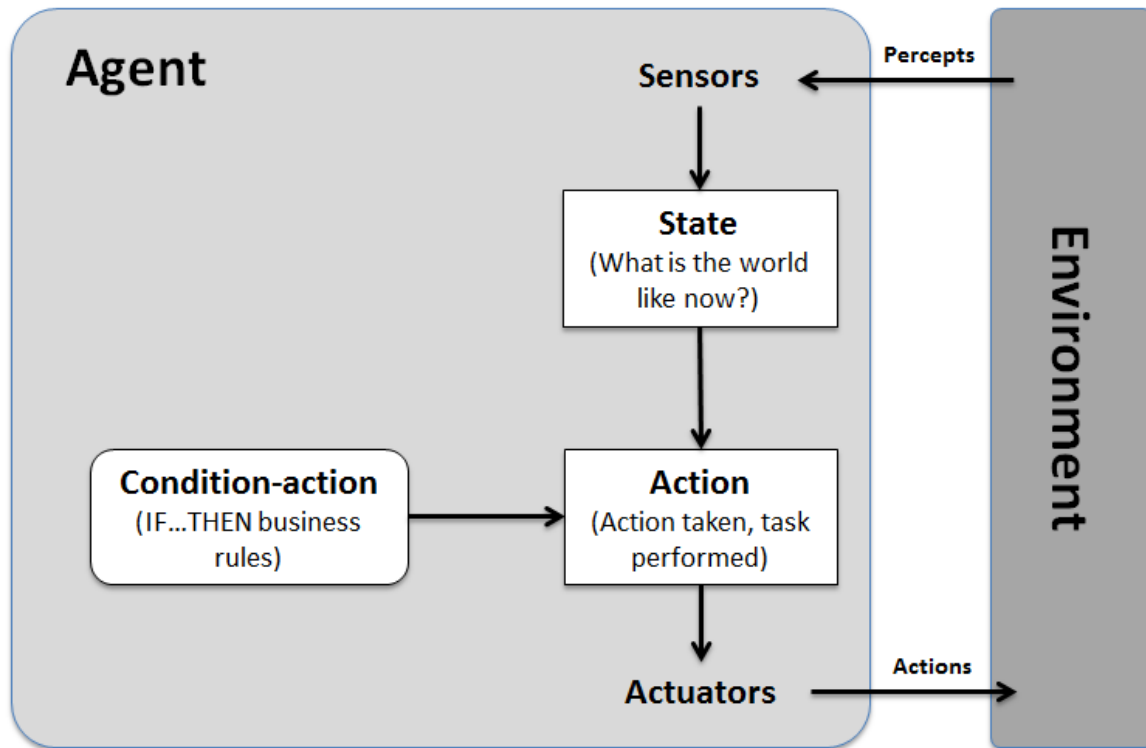
An **agent** is an entity capable of **sensing** the **state** of its **environment** and **acting** upon it based on a set of specified **rules**. An agent performs specific tasks on behalf of another. In the case of software, an agent is a software program. There are many different types of intelligent software agents¹⁰⁹.

¹⁰⁶ Blackline, Accounting Process Automation, <https://www.blackline.com/accounting-process-automation>

¹⁰⁷ FlexRule, Business Logic Platform, <http://www.flexrule.com/solution/>

¹⁰⁸ *Comprehensive Introduction to Intelligent Software Agents*, http://xbrlsite.azurewebsites.net/2017/IntelligentDigitalFinancialReporting/Part01_Chapter02.7_ComprehensiveIntroductionToIntelligentSoftwareAgents.pdf

¹⁰⁹ Wikipedia, *Intelligent Agent*, Retrieved July 24, 2016; https://en.wikipedia.org/wiki/Intelligent_agent



A simple reflex agent looks up what it should do from a **list of rules** in response to its perception to the environment. This is the algorithm of a simple reflex agent¹¹⁰:

Function Simple-Reflex-Agent (percept) returns action

- **persistent:** *rules*, a set of condition-action rules
- *state* <<<< Interpret-Input (*percept*)
- *rule* <<<< (Rule-Match (*state*, *rules*))
- *action* <<<< *rule*. Action
- **return** *action*

The document *Comprehensive Introduction to Intelligent Software Agents*¹¹¹ goes into significantly more detail on the topic of intelligent software agents.

Categories of Intelligent Software Agent Functionality

The functionality of an intelligent software agent can be classified into groups, or categories:

¹¹⁰ Hugo Larochelle, *Intelligence Artificielle*, Retrieved July 24, 2016; <https://www.youtube.com/watch?v=TUHAVbaBLIg>

¹¹¹ Charles Hoffman and Rene van Egmond, *Comprehensive Introduction to Intelligence Software Agents*, http://xbrlsite.azurewebsites.net/2017/IntelligentDigitalFinancialReporting/Part01_Chapter02.7_ComprehensiveIntroductionToIntelligentSoftwareAgents.pdf

- **Reactive agent:** A reactive agent is capable of maintaining an ongoing interaction with the environment and responding in a timely fashion to changes that occur in it.
- **Pro-active agent:** A pro-active agent is capable of taking the initiative; not driven solely by events, but capable of generating goals and acting rationally to achieve them.
- **Deliberative agent:** A deliberative agent symbolically represents knowledge and makes use of mental notions such as beliefs, intentions, desires, choices and so on. This is implemented using a belief-desire-intension model.
- **Hybrid agent:** A hybrid agent is one that mixes some of all the different architectures.

Every intelligent software agent falls into one of those four categories.

Categories of Intelligent Software Agent Sophistication

Intelligent software agents can be grouped as to the level of sophistication offered by the agent:

- **Generic agent:** An agent is anything that perceives an environment through sensors and acts or reacts upon the environment through effectors.
- **Simple reflex agent:** A simple reflex agent looks up what it should do from a list of rules in response to its perception to the environment.
- **Model-based reflex agent:** A model-based reflex agent is the same thing as a simple reflex agent except that a model-based reflex agent has a model of how the environment evolves.
- **Goal-based agent:** A goal-based agent has a goal or set of goals that it actively pursues in accordance with an agenda (so this type of agent is proactive, not just reactive). A goal based agent has a representation of the current state of the environment and how that environment works. The agent pursues policies or goals that may not be immediately attainable. And so, goal based agents do not live merely in the moment. These agents consider different scenarios before acting on their environments, to see which action will probably attain a goal. This consideration of different scenarios is called search and planning.
- **Utility-based agent:** A utility-based agent is a more sophisticated type of goal-based agent that also rates each possible scenario to see how well it achieves certain criteria with regard to production of the good outcome, therefore it is more adaptive. A utility measure is applied to the different possible actions that can be performed in the environment. The utility-based agent will rate each scenario to see how well it achieves certain criteria with regard to the production of a good outcome. Things like the probability of success, the resources needed to execute the scenario, the importance of the goal to be achieved, the time it will take, might all be factored in to the utility function calculations.

Every intelligent software agent falls into one of those groups in terms of sophistication of functionality offered.

Benefits of an Expert System

Benefits from the use of expert systems include:

- **Reduced costs by using automation:** elimination of routine, boring, repetitive, mundane, mechanical, rote tasks that can be automated
- **Increased uniformity:** consistent answers from the same question or facts; computers are good at performing repetitive, mechanical tasks whereas humans are not; computers do not make mistakes and are good at repeating exactly the same thing each time; performance level is consistent
- **Reduced down-time:** computer based expert systems are tireless and do not get distracted
- **Increased availability:** computer based expert systems are always available simultaneously in multiple places at one time; you get quick response times and can replace absent or scarce experts; convenient
- **Diligence and tenacity:** computers excel at paying attention to detail; they never get bored or overwhelmed and they are always available and will keep doing their job until the task is complete with the same attention to detail
- **Basis for training:** the best practices of the best practitioners can be available to those that are new to and learning about a domain of knowledge
- **Longevity and persistence:** computer based expert systems do not change jobs or retire so knowledge gathered by an organization can remain within that organization
- **Productivity:** computer based expert systems are cheaper than hiring experts and costs can be reduced at the same time that quality increases resulting in increased productivity
- **Multiple opinions:** Systems can integrate the view of multiple experts within a system and choose between the preferred view of multiple expert opinions in the same system
- **Objectivity:** computers apply the same inductive and deductive logic consistently; emotion and personal preferences can be eliminated where they should be eliminated; expert systems do not discriminate
- **Easier dissemination of knowledge:** expert systems are software and metadata and therefore once you have that software and metadata reproducing another version is trivial and the incremental cost is extremely low

In a knowledge based or expert system; knowledge is explicitly represented and can be evaluated, knowledge is permanent, knowledge is easily replicated, and the system is consistent. Operating costs of an expert system are low. Financial report creation software of

the future will be an expert system which operates similar to how CAD/CAM software for creating blueprints.

Disadvantages of Expert Systems

Everything has advantages and disadvantages. The following can be disadvantages of expert systems:

- **Initial cost:** the initial cost of creating an expert system can be high; the primary cost is for creation of the expert knowledge which is used by the system
- **Maintaining knowledge:** human experts constantly update their knowledge through interaction with other experts, new ideas, common sense, etc.; expert systems have to be maintained to keep knowledge current
- **Garbage in, garbage out:** an expert system is only as good as the machine-readable knowledge which the system uses
- **No common sense:** humans have common sense, expert systems do not
- **Lacks human touch:** expert systems are computer application and have the same characteristics of a computer; they have no compassion, no intuition, cannot exercise real judgment, etc.
- **Inflexibility:** a system, once set up, is inflexible or rather only flexible to the extent that new knowledge is added to the system
- **Restricted:** an expert system usually has expertise in one specific domain of knowledge and is therefore restricted to that specific knowledge

To make expert systems work effectively, disadvantages must be overcome and expert systems should be used to solve problems they are truly capable of solving. Setting the right expectations is important.

Understanding the Benefits of Double-entry Accounting

Single-entry bookkeeping¹¹² is how 'everyone' would do accounting. In fact, that is how accounting was done before double-entry bookkeeping was invented.

Double-entry bookkeeping¹¹³ adds an additional important property to the accounting system, that of a clear strategy to identify errors and to remove them from the system. Even better, it has a side effect of clearly firewalling errors as either accident or fraud. This then leads to an audit strategy. Double-entry bookkeeping is how professional accountants do accounting.

¹¹² Wikipedia, *Single-entry Bookkeeping System*, retrieved August 30, 2016, https://en.wikipedia.org/wiki/Single-entry_bookkeeping_system

¹¹³ Wikipedia, *Double-entry Bookkeeping System*, retrieved August 30, 2016, https://en.wikipedia.org/wiki/Double-entry_bookkeeping_system

Double-entry bookkeeping, documented by the Italian mathematician and Franciscan Friar Luca Pacioli¹¹⁴, is one of the greatest discoveries of commerce, and its significance is difficult to overstate.

Which came first, double-entry bookkeeping or the enterprise¹¹⁵? Was it double-entry bookkeeping and what it offered that enable the large enterprise to exist; or did the large enterprise create the need for double-entry bookkeeping?

Accountants think differently than non-accountants. Non-accountants don't realize this and accountants seem to forget. The quality difference between the set of "stuff" that makes up a financial report and all the support for that financial report tends to be much higher than the quality level of non-financial information, or rather, information that is managed by a non-accountant. Why? Because double-entry bookkeeping is ingrained in the processes and procedures of accountants.

Accountants need to keep this in mind as they design digital financial reporting.

Logic

As was pointed out earlier, a logical theory defines and describes things with clarity. Details about something like a financial report must be explained to brainless robots. Once you define things you can ask questions. Further, the discipline of describing something in a form a computer algorithm can understand also assists you in understanding the world better; weeding out flaws in your understanding, myths and misconceptions.

Logic a set of principles that form a framework for correct reasoning. Logic is a process of deducing information correctly. Logic is about the correct methods that can be used to prove a statement is true or false. Logic tells us exactly what is meant. Logic allows systems to be proven.

The principles of logic are topic-neutral, universal principles which are more general than say the single domain of biology, mathematics, accounting, or economics. Logic has to do with the meaning of concepts common to all domains and establishes general rules governing concepts. Logical truths are necessary. The principles of logic are derived solely using reasoning and the validity of the principles are not dependent on any other feature of the world.

Logic is the process of deducing information correctly; logic is not about deducing correct information. Understanding the distinction between **correct logic** and **correct information** is

¹¹⁴ Wikipedia, *Luca Pacioli*, retrieved August 30, 2016, https://en.wikipedia.org/wiki/Luca_Pacioli

¹¹⁵ Ian Grigg, *Triple Entry Accounting, A Very Brief History of Accounting, Which Came First - Double Entry or the Enterprise?*, http://iang.org/papers/triple_entry.html

important because it is important to follow the consequences of an incorrect assumption. Ideally, we want both our logic to be correct and the facts we are applying the logic to, to be correct. But the point here is that correct logic and correct information are two different things. If our logic is correct, then anything we deduce from such information will also be correct.

Logic is also not the *psychology of reasoning*¹¹⁶. Again, logic tells us how we ought to reason if we want to reason correctly. Whether people follow these rules is not our concern here. The psychology of reasoning helps one understand the actual reasoning habits of people including reasoning mistakes. Again, our concern is getting the logic correct.

Business professionals have an innate understanding of logic. This understanding can be employed to make it easier to work with machines.

Logic is a system, process, and principles of correct reasoning. Logic is about deducing information correctly. Another term that is similar to logic is critical thinking. **Critical thinking** is defined as¹¹⁷:

Critical thinking is the intellectually disciplined process of actively and skilfully conceptualizing, applying, analysing, synthesizing, and/or evaluating information gathered from, or generated by, observation, experience, reflection, reasoning, or communication, as a guide to belief and action.

Yet another term which seems to be quite similar to logic and critical thinking is computational thinking. **Computational thinking** is defined as¹¹⁸:

Computational thinking is a thought process involved in formulating problems and their solutions so that the solutions are represented in a logical, clear, and systematic form that can be explained to and effectively carried out by a computer or human.

The primary thing that is studied in logic is the principles governing the validity of statements (i.e. reasoning) and whether certain conclusions (i.e. deductions) follow from some given facts and rules per the principles of logic.

Whatever term you might prefer the bottom line is that logic, critical thinking, and/or computational thinking all tools to help you to (a) build computer systems right and (b) build the right computer systems. After all, computers work using the rules of logic.

¹¹⁶ University of Hong Kong, *What is Logic*, §1. *Logic is not the psychology of reasoning*, <https://philosophy.hku.hk/think/logic/whatislogic.php>

¹¹⁷ The Foundation for Critical Thinking, *Defining Critical Thinking*, <https://www.criticalthinking.org/pages/defining-critical-thinking/766>

¹¹⁸ Computational Thinking is for Everyone, <http://xbrl.squarespace.com/journal/2018/9/22/computational-thinking-is-for-everyone.html>

Difference between Assuming and Deriving Information

Assuming new information and logically deriving new information from other information is not the same thing. When one make an assumption¹¹⁹ you “take it upon oneself” or “lay claim to” or “take for granted” some fact or statement as being true. When you derive new information using logical processes however, there is a logically supportable or logically defensible line of reasoning that (a) follow the rules of logic and (b) result in some new but supportable true fact or statement.

Fundamentals of Logic

Professional accountants use logic informally every day. For example, something that you probably learned in school is BASE; beginning balance + additions – subtractions = ending balance. If you know any three facts, you can always find the forth fact.

Here is an example of logic. Suppose you were trying to find the subtractions from some account.

1. You know that the *beginning balance* of the account is \$5,000
2. You know that the *additions* to the account was \$1,000
3. You know the *ending balance* of the account is \$2,000
4. You know that beginning balance + additions – subtractions = ending balance
5. You can derive the fact that the *subtractions* to the account are \$4,000 using the information provided in #1, #2, #3, and #4 and using the rules of logic

In this example you are using logic to reliably derive the subtractions from some account even though you do not know the actual value of the subtractions. And in doing so you are adding new information to your base of knowledge.

There is an important point to recognize here. Logic is the process of deducing information correctly; logic is not about deducing correct information. Consider the example above again. Suppose you were give incorrect information for the beginning balance of the account you are analyzing, that the beginning balance was \$9,000 rather than \$5,000 as stated above.

Our deduction that the subtractions are \$4,000 is now untrue. But the logic is perfectly correct; the information was pieced together correctly, even if some of that information was false.

Understanding the distinction between correct logic and correct information is important because it is important to follow the consequences of an incorrect assumption.

¹¹⁹ Merriam-Webster, Assumption, <https://www.merriam-webster.com/dictionary/assumption>

Ideally, we want both our logic to be correct and the facts we are applying the logic to, to be correct. But the point here is that correct logic and correct information are two different things.

If our logic is correct, then anything we deduce from such information will also be correct.

Logical Statements

In logic, a **statement** is a sentence that is either true or false. You can think of statements as pieces of information that are either correct or incorrect. And therefore, statements are pieces of information that you apply logic to in order to derive other pieces of information which are also statements. Statements are basically rules.

A **logical theory** is a set of logical statements that formally describes some subject or system. **Axioms**¹²⁰ are statements that describe self-evident logical principles that no one would argue with. **Theorems**¹²¹ are logical deductions which can be proven by constructing a chain of reasoning by applying axioms and the rules of logic in the form of IF...THEN statements.

A **rule**, or business rule or assertion, is a true statement with respect to some model of the real world that could possibly exist given some logical theory. You cannot create rules that are true in worlds that can never exist. A rule can be a mathematical expression. A rule is a type of logical statement.

Here are some examples of statements:

- “Assets = Liabilities and equity”, i.e. the accounting equation¹²².
- “Beginning balance + additions – subtractions = ending balance”, i.e. as stated above and the definition of a roll forward.
- “Originally stated balance + adjustments = Restated balance”
- Some types of revenue are operating.
- Assets for the consolidated entity as of the balance sheet date of December 31, 2016 is \$45,000.

A financial report discloses facts. One fact in a report is distinguishable from another fact in the report by the characteristics of each of the two facts. One fact in a report can be related to another fact in a report and the relation between those two facts should be consistent with expectations which are established by rules.

¹²⁰ Wikipedia, *Axiom*, <https://en.wikipedia.org/wiki/Axiom>

¹²¹ Wikipedia, *Theorem*, <https://en.wikipedia.org/wiki/Theorem>

¹²² Wikipedia, *Accounting Equation*, retrieved June 2, 2017, https://en.wikipedia.org/wiki/Accounting_equation

Logic is about the correct methods that can be used to prove that a statement is true or false. To prove that a statement is true, we start with statements other statements that are proven to be true and use logic to deduce more and more complex statements until finally we obtain a statement that we are looking to determine if the statement is true or false. Of course some statements are more difficult to prove than others; in this resource we will concentrate on statements that are easier to prove to help you learn the basics of logic. But the point is this: In proving that statements are true, we use logic to help us understand statements and to combine pieces of information to produce new pieces of information.

As your skills grow you can create increasing complex statements.

A financial report is complex logical information¹²³. XBRL-based financial reports are machine-readable structured information.

Conditional Statements

Another way to represent statements is in the IF...THEN format. These are called conditional statements. Here is an example of a conditional statement¹²⁴.

- IF Assets = Liabilities and Equity; THEN the balance sheet balances.

Note that you may not be able to reverse a conditional statement. For example, the above conditional statement reversed would be:

- IF the balance sheet balances; THEN Assets = Liabilities and Equity.

The “IF” part of a conditional statement is called the *hypothesis* and the “THEN” part is called a *conclusion*. There are other terms that are used such as the *antecedent* and *consequent*.

Logical Inference

Logical inference is the process of deriving new information from one or more existing pieces of information, deducing a conclusion about that information using the rules of logic. For example,

- Suppose you know that Assets = \$2,000
- Suppose you know that Current assets = \$500
- Suppose you know that Assets = Current assets + Noncurrent assets

Using the information provided above, you can use the rules of logic to derive the value of Noncurrent assets to be \$1,500 because Assets (\$2,000) = Current assets (\$500) + Noncurrent

¹²³ Charles Hoffman, *Processing Complex Logical Information or Structured Knowledge*, <http://xbrl.squarespace.com/journal/2017/4/22/processing-complex-logical-information-or-structured-knowled.html>

¹²⁴ Wikipedia, *Material Conditional*, retrieved June 2, 2017, https://en.wikipedia.org/wiki/Material_conditional

assets (UNKNOWN); but using the rules of math you solve for the value of the UNKNOWN;
Assets (\$2,000) – Current Assets (\$500) = Noncurrent assets (UNKNOWN); finally you get to
Noncurrent assets = \$1,500.

You are not guessing. You are using logic to determine, accurately, what the value of
Noncurrent Assets is based on other facts that you know.

Contrasting Deductive Reasoning and Inductive Reasoning

Artificial intelligence has an issue¹²⁵. You need to be aware of this issue because it means that
certain types of artificial intelligence should not be employed for solving certain classes of
problems. The issue is straight forward.

There are two primary types of reasoning systems¹²⁶: **deduction**¹²⁷ and **induction**¹²⁸. These two
videos help you understand the difference between deduction and induction: *What is*
*Deduction and Induction*¹²⁹; *What is Inductive Logic*¹³⁰.

The bottom line is that *deduction* is certain; you can be logically certain of the conclusions
reached using deduction. That is the goal of deduction, certainty. *Induction* is not certain,
induction is based on probability.

The black swan problem of induction¹³¹ clearly explains the problem of induction. “No amount
of observations of white swans can allow the inference that all swans are white, but the
observation of a single black swan is sufficient to refute that conclusion.”

Many employing artificial intelligence use inductive reasoning systems. They believe that these
systems can work because the problems caused by induction can be managed effectively. That
may be true. But it likewise may not be true for certain knowledge domains. The call here
should be made by domain experts, not computer scientists that do not have a deep
understanding of the knowledge domain.

Setting the Right Expectation

Logic has limitations¹³². Business professionals need to understand these limitations so that
they understand what computers can and cannot do, what is hard and what is easy to

¹²⁵ For a deep analysis see, Peter Sweeney, *One problem to explain why AI works*, <https://towardsdatascience.com/one-problem-to-explain-ai-218a29e8fbc0>

¹²⁶ Wikipedia, *Reasoning System*, https://en.wikipedia.org/wiki/Reasoning_system

¹²⁷ Wikipedia, *Deductive Reasoning*, https://en.wikipedia.org/wiki/Deductive_reasoning

¹²⁸ Wikipedia, *Inductive Reasoning*, https://en.wikipedia.org/wiki/Inductive_reasoning

¹²⁹ YouTube.com, *What are Deduction & Induction?*, <https://www.youtube.com/watch?v=iRcNQkWNWNk>

¹³⁰ YouTube.com, *What is Inductive Logic?*, <https://www.youtube.com/watch?v=4ZKa1S1wPy4>

¹³¹ Capture, *Black Swans and the Problem of Induction*, <https://capturehighered.com/predictive-modeling/black-swans-problem-induction/>

¹³² Martin Kuba, Institute of Computer Science, OWL 2 and SWRL Tutorial, *Limitations of First-order logic expressiveness*,
<http://dior.ics.muni.cz/~makub/owl/>

implement using computers, and to otherwise set their expectations appropriately. Remember, computers cannot perform magic. Computers fundamentally follow the rules of mathematics which follow the rules of logic. It really is that straight forward.

It is more difficult, but not impossible, to get computers to effectively work with information such as the following:

- fuzzy expressions: “It **often** rains in autumn.”
- non-monotonicity: “Birds fly, penguin is a bird, but a penguin does not fly.”
- propositional attitudes: “Eve **thinks** that 2 is not a prime number.” (It is true that she thinks it, but what she thinks is not true.)
- modal logic
 - possibility and necessity: “It is **possible** that it will rain today.”
 - epistemic modalities: “Eve **knows** that 2 is a prime number.”
 - temporal logic: “I am **always** hungry.”
 - deontic logic: “You **must** do this.”

While it is possible to implement this sort of functionality within computer systems using technologies such as probabilistic reasoning¹³³, those systems will be less reliable and significantly more difficult to create. On the other hand, probabilistic reasoning can provide value. The bottom line is this: what are the effective boundaries of the system?

Digital Mirror Worlds Representing Reality

In their paper *Imagineering Audit 4.0*¹³⁴, Jun Dai and Miklos Vasarhelyi of Rutgers University use the term **mirror world** to describe the use of technology to create a virtual copy of the real world. A mirror world is defined as an “informationally-enhanced virtual models or reflections of the physical world.”

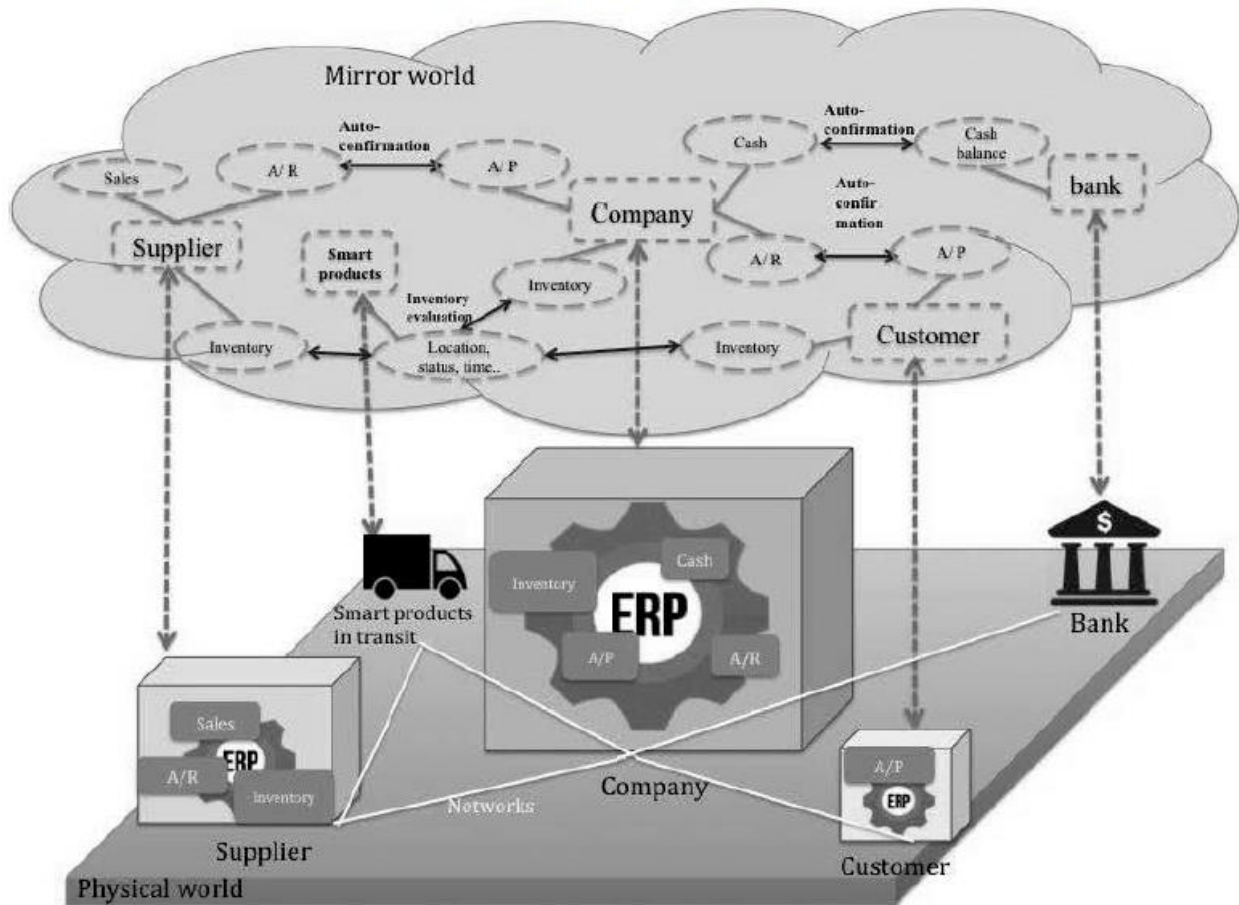
Figure 3 on page 10 of Dai and Vasarhelyi’s paper provides a graphic¹³⁵ describes the relationships between inter-business parties in a supply chain and represents the connections between the real world in which such businesses operate and the mirror world within the accounting processes of an enterprise resource planning (ERP) system of those economic entities.

¹³³ Wikipedia, *Probabilistic Logic*, retrieved August 28, 2016, https://en.wikipedia.org/wiki/Probabilistic_logic

¹³⁴ Jun Dai and Miklos Vasarhelyi, *Imagineering Audit 4.0*, <http://aaajournals.org/doi/abs/10.2308/jeta-10494>

¹³⁵ Jun Dai and Miklos Vasarhelyi, *Imagineering Audit 4.0*, figure 3 on page 10, <http://aaajournals.org/doi/abs/10.2308/jeta-10494>

FIGURE 3
Audit 4.0 of Inter-Business Parties



The extent that the real world can be represented within some mirror world can be debated. However, what cannot be debated is that the real world and the mirror world need to use the same logic to explain such worlds. (i.e. different systems cannot use different or irreconcilable logic to represent the same world).

Another term for the notion of a digital mirror world is “**digital twin**”. A digital twin is defined as, “Digital twin is the phrase used to describe a computerized (or digital) version of a physical asset and/or process. The digital twin contains one or more sensors that collect data to represent real-time information about the physical asset.”¹³⁶

¹³⁶ Webopedia, Digital Twin, <https://www.webopedia.com/TERM/D/digital-twin.html>

But, Which Logic?

The *Internet Encyclopedia of Philosophy*¹³⁷ describes logic as follows, “The aim of logic is the elaboration of a coherent system that allows us to investigate, classify, and evaluate good and bad forms of reasoning.”

It appears that there are two primary forms of logic: traditional logic and modern mathematical logic¹³⁸. It appears that there are some differences between traditional and modern mathematical logic. One difference is that modern mathematical logic introduced the truth table. People seem to disagree on the differences between traditional logic and modern mathematical logic¹³⁹. Martin Cothran explains the difference in *The Difference between Traditional and Modern Logic and the Difference it Makes*¹⁴⁰ thus:

“First, the assumptions behind the two systems are different. Second, the format or structure of the two systems is different. Third, the respective purposes of the two systems are different. In all three cases—in the assumptions, structure, and the purpose—the traditional system reflects traditional views, and the modern system reflects modern views about reality. Each system is based on a different metaphysic.”

Practical Logic for Business Professionals

I am going to create my own logic framework. I am going to call it “unifying practical logic framework for business professionals¹⁴¹”. My logic is grounded in both traditional logic and modern logic, taking the best of each, and will have clear boundaries. Here are my requirements:

1. The system of logic MUST be understandable to the average business professional.
2. The system of logic MUST be consistent with first-order logic and modern mathematical logic.
3. The goal of the system is to enable precise communications between business professionals, information technology professionals, and knowledge engineers.

I don’t have this all worked out yet, but this is what I have thus far:

¹³⁷ Internet Encyclopedia of Philosophy, *Aristotle:Logic*, <http://www.iep.utm.edu/aris-log/>

¹³⁸ Martin Cothran, *The Difference between Traditional and Modern Logic and the Difference it Makes*, <http://vereloqui.blogspot.com/2012/12/the-difference-between-traditional-and.html>

¹³⁹ Wilfrid Hodges, *Traditional versus Modern Logic*, <http://wilfridhodes.co.uk/history17.pdf>

¹⁴⁰ Martin Cothran, *The Difference between Traditional and Modern Logic and the Difference it Makes*, <http://vereloqui.blogspot.com/2012/12/the-difference-between-traditional-and.html>

¹⁴¹ Charles Hoffman, CPA, *Unifying Logic Framework for Business*, <http://xbrl.squarespace.com/journal/2017/11/26/unifying-logic-framework-for-business.html>

There are two fundamental underlying assumptions that serve as basic building blocks.

Assumption 1: A statement **MUST** be either true or false. **Assumption 2:** A statement cannot be both true and false at the same time in the same context.

These are the basic definitions:

- **Logic** a set of principles that forms a framework for correct reasoning. Logic is a process of deducing information correctly. Logic is about the correct methods that can be used to prove a statement is true or false. Logic tells us exactly what is meant. Logic allows systems to be proven.
- A logical **statement**, or a statement of fact, is a sentence that carries factual information that is either true or false. A statement is a piece of information. A **conditional statement** is a type of statement that has an IF...THEN type format. Look at a statement as being a piece of information that is either correct or incorrect. A statement, a statement of fact, and a fact mean the same thing. Questions, commands, and opinions are not logical statements.
- A **model**, or conceptual model, describes a possible world. There exists some set of all possible models that can be used to describe real worlds that could exist.
- A **rule**, or business rule or assertion, is a true statement with respect to some model of the real world that could possibly exist. (i.e. you cannot create rules that are true in worlds that can never exist) A rule can be a mathematical expression. A rule is a type of logical statement.
- **Logical entailment**, or logical consequent, is when a logical statement follows from another statement or set of statements. Another name for logical entailment is inference. The rules of inference provide a system in which we can produce new information (statements) from known information (statements).
- A statement has a **truth value** with respect to some model.
- The **connectors AND, OR, and NOT** are used to combined statements to create compound statements.

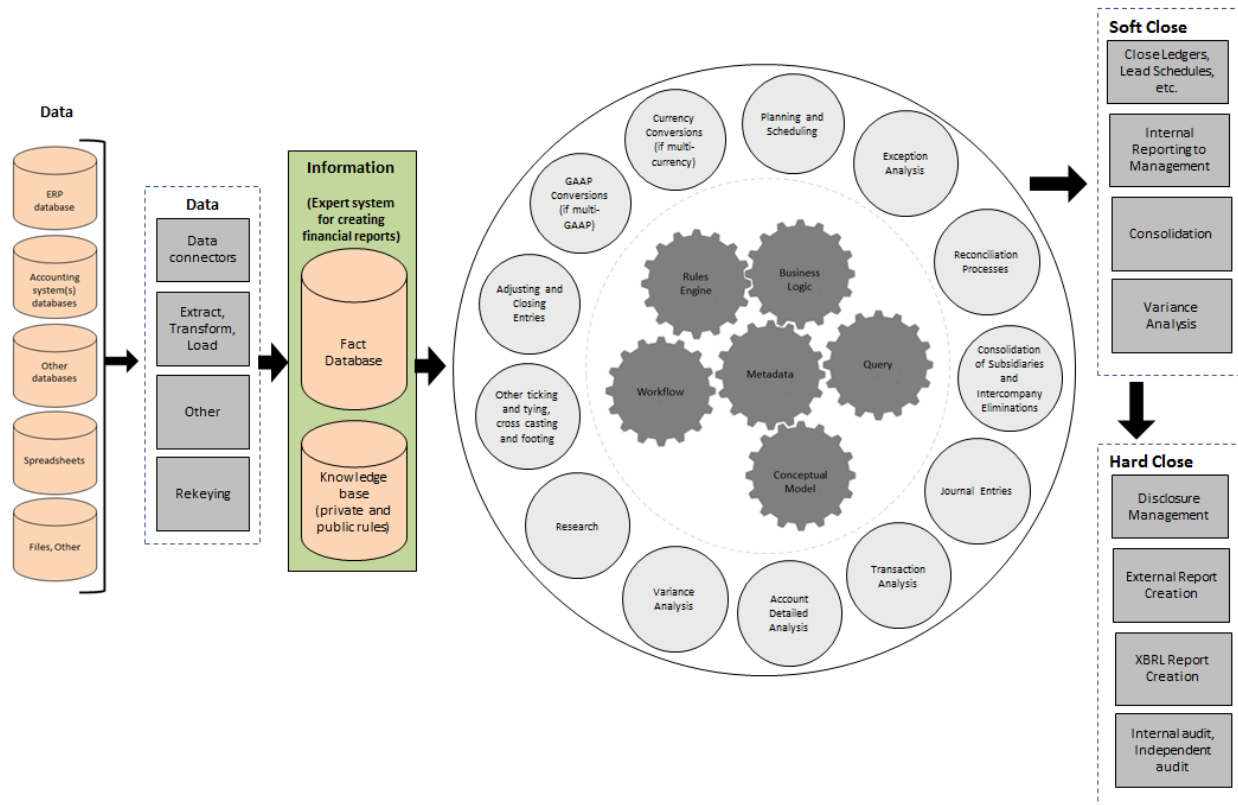
Logic helps us understand the meaning of statements and produce new meaningful statements. Logic is the glue that holds strings of statements together and pins down the exact true unambiguous meaning of those statements.

Logic is the process of deducing information correctly; logic is not about deducing correct information. Suppose you were give incorrect information. Your statements can be false but the reasoning behind conclusion sound. Other means are necessary to determine if a statement is true or false.

Tasks

Facts, rules, and the possible models they represent can be stored in a **knowledge based system** if the facts and rules are put into machine-readable form. The machine-readable system can apply a **problem solving logic** using a **problem solving method** of a **business rules engine** to solve problems that normally would require human effort and thought to solve. The knowledge based system supplies an **explanation and justification mechanism** to help system users to understand the **line of reasoning** used and support conclusions reached by the machine-based knowledge based system, presents that information to the user of the system, and provide an audit trail. (i.e. how the system works cannot be a black box).

The fact database and knowledge base can be employed to effectively automate certain tasks involved in accounting processes and financial report creation processes.



The knowledge based system keeps track of things. You can **TELL** the knowledge based system facts or rules; you can **ASK** for facts that exist in the knowledge based system or new facts that can be inferred using the rules of logic, existing facts, and existing rules. For example,

- **TELL** statement “Current assets for the economic entity ABC Company as of December 31, 2017 is the amount \$200,000 (USD).”
- **TELL** statement “Assets for the economic entity ABC Company as of December 31, 2017 is the amount \$1,000,000 (USD).”

- **TELL** statement “For commercial and industrial companies; Assets = Current Assets + Noncurrent Assets.”
- **TELL** statement “ABC Company is a commercial and industrial company”.
- **ASK** fact “What is the amount of Noncurrent Assets for ABC Company as of December 31, 2017 in USD?”

The notions of TELLING and ASKING are guided by the rules of logic. To understand the importance of these rules of logic, think about how such a system would work without such rules. The result would be anarchy.

It is the rules of logic and the ability to represent facts and rules in machine-readable form but also in human-readable form that can enable the creation of a digital mirror world that can be used to automate accounting and reporting processes. Some call this “robotic finance”¹⁴². Others call this machine intelligence¹⁴³. Still others refer to this as the artificial intelligence revolution¹⁴⁴.

It is important to remember that logic is the process of deducing information correctly; **logic is not about deducing correct information**. Deciding which information is correct and which is not correct is the responsibility of the model you choose to make use of to describe your reality.

Digital mirror worlds can be created that leverage the nature of double-entry bookkeeping¹⁴⁵ using the XBRL technical syntax¹⁴⁶. To do this, one needs to consider computer empathy to understand how to do this and the extent that accounting process automation can realistically be achieved.

Final Thoughts about Logic

Logic was invented by Aristotle in the 4th century BCE. Logic is a discipline of philosophy and can serve as a Rosetta Stone that helps business professionals communicate with information technology professionals and knowledge engineers¹⁴⁷. Logic is a common language that can be agreed upon, understood by all parties, and which therefore enables precise communication.

¹⁴² Robert Kugel, Ventana Research, *Welcome to the Age of Robotic Finance*, <http://xbrl.squarespace.com/journal/2018/3/19/ventura-research-welcome-to-the-age-of-robotic-finance.html>

¹⁴³ Shivon Zilis and James Cham, Harvard Business Review, *The Competitive Landscape for Machine Intelligence*, <https://hbr.org/2016/11/the-competitive-landscape-for-machine-intelligence>

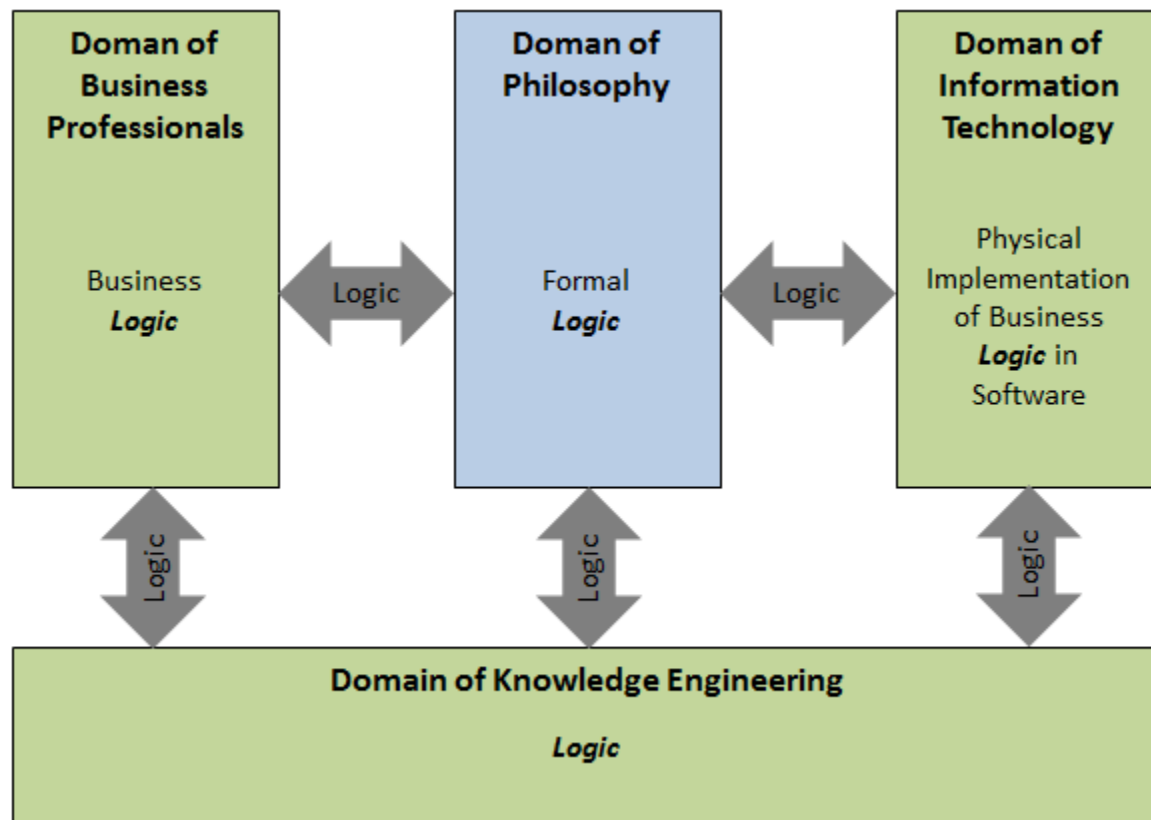
¹⁴⁴ Ken Tysiac, Journal of Accountancy, *Accountants role in managing AI disruption*, <https://www.journalofaccountancy.com/news/2017/oct/accountants-role-in-managing-artificial-intelligence-disruption-201717688.html>

¹⁴⁵ Charles Hoffman, CPA, *Accounting Process Automation Using XBRL*, <http://xbrl.azurewebsites.net/2018/Library/AccountingProcessAutomationUsingXBRL.pdf>

¹⁴⁶ Charles Hoffman, CPA, *Guide to Implementing Robotic Finance*, <http://xbrl.azurewebsites.net/2018/RoboticFinance/GuideToImplementingRoboticFinance.pdf>

¹⁴⁷ Ronald Fuller, *Business and IT Professionals Should Study Formal Logic*, <http://oninfo.info/11-information/56-business-and-it-professionals-should-study-formal-logic>

Both computer science and philosophy are concerned with the representation of information and rational inference. Philosophy enhances the kind of thinking required to be clear, analytical, and concise. Business domains are concerned with the representation of information and rational inference because they have information they need to represent.



Logic helps you get to true and unambiguous meaning and helps you reconcile your true and unambiguous meaning to the true and unambiguous meaning of others. Computers are machines that work using the rules of logic. Logic helps to keep everyone on the same page.

It is important to understand why you want to take the time to understand logic. Here are the primary reasons:

- Applying logic correctly helps you understand what something means.
- The rules of inference provide a system which can be used to reliably derive new information from existing information.
- Logic helps you not only understand the meaning of statements; it also helps you reliably produce new meaningful statements.
- Computers work based on the rules of mathematics. Mathematics works based on the rules of logic.

Controlling Process Quality

Accounting is a process. A kludge (or kluge) is an engineering/computer science term that describes what is best described as a workaround or quick-and-dirty solution that is typically clumsy, inelegant, inefficient, difficult to extend and hard to maintain; but it gets the job done. By contrast, elegance is beauty that shows unusual effectiveness and simplicity.

Some accounting systems and their related processes are kludges that are cobbled together over many years. Some accounting systems and processes are well-engineered. Accountants sometimes spend a lot of time improving the quality of things that should be destroyed.

Idealized Redesign

Idealized redesign is the notion of imagining what you would do to improve some process or system if you have no constraints. What would you do to improve your system if you were unconstrained? If you cannot answer that question and improve a system when unconstrained; you certainly cannot improve a system given the realities of constraints that you must generally live with. Another term for idealized redesign is greenfield project¹⁴⁸.

Lean

Think about something. In your 40 to 60 hour work week, how much time do you get to think? How much time do you get to be proactive? To be creative? To apply what you have been trained to do¹⁴⁹?

A process is about satisfying some customer. Lean process methodology is about removing tasks that don't add value and (b) looking carefully to find waste and remove that waste from a system.

Six Sigma

Six Sigma is a set of techniques and tools for process improvement. Six Sigma is about measurement. Six Sigma essentially targets enhancing the efficiency of business processes by minimizing the number of defects output by processes. The goal is to attain near-perfect processes.

If you can't measure it, you can't control it (i.e. if you can measure it, you can control it). The goal of Six Sigma is to have 3.4 errors per 10,000.

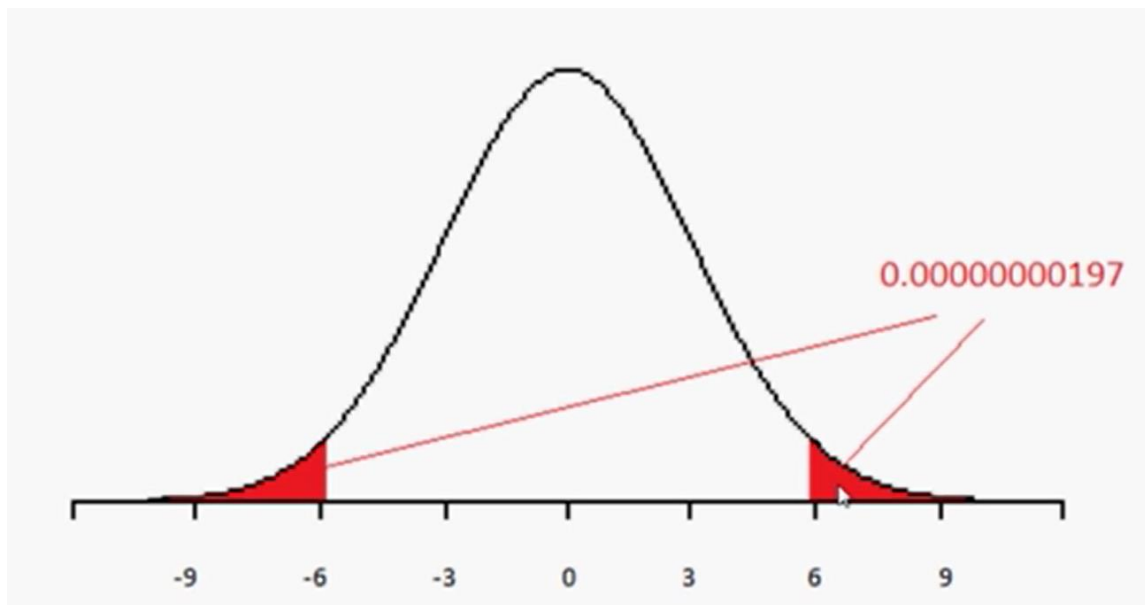
¹⁴⁸ Wikipedia, *Greenfield Project*, https://en.wikipedia.org/wiki/Greenfield_project

¹⁴⁹ YouTube.com, *Bill Peterson - Lean Applied to Us*, <https://www.youtube.com/watch?v=tfQiGDUBdD0>



Sigma level	Sigma (with 1.5 σ shift)	DPMO	Percent defective	Percentage yield	Short-term C_{pk}	Long-term C_{pk}
1	-0.5	691,462	69%	31%	0.33	-0.17
2	0.5	308,538	31%	69%	0.67	0.17
3	1.5	66,807	6.7%	93.3%	1.00	0.5
4	2.5	6,210	0.62%	99.38%	1.33	0.83
5	3.5	233	0.023%	99.977%	1.67	1.17
6	4.5	3.4	0.00034%	99.99966%	2.00	1.5
7	5.5	0.019	0.0000019%	99.9999981%	2.33	1.83

Six Sigma is about the proactive control of boundaries. Set upper boundaries and lower boundaries such that quality control will be alerted and corrective action taken before problems occur.



Lean Six Sigma

Born in manufacturing, Lean Six Sigma is just as applicable to accounting processes and accounting information systems as it is to building a physical product.

Lean Six Sigma¹⁵⁰ is a discipline that combines the problem solving methodologies and quality enhancement techniques of Six Sigma¹⁵¹ with the process improvement tools and efficiency concepts of Lean Manufacturing¹⁵². Born in the manufacturing sector, Lean Six Sigma works to produce products and services in a way that meets consumer demand without creating wasted time, money and resources.

Specifically, Lean is ‘the purposeful elimination of wasteful activities.’ It focuses on making process throughout your company faster, which effects production over a period of time. Six Sigma works to develop a measurable process that is nearly flawless in terms of defects, while improving quality and removing as much variation as possible from the system.

Defining Quality

There are many different ways to define or think about quality. Engineer and statistician W. Edwards Deming¹⁵³ defined quality as “predictability,” and called variance “the enemy of quality.” To achieve an intended outcome, Deming thought it was important to plan for common-cause variation, which can be predicted, and special-cause variation, which cannot be predicted.

Harold F. Dodge, one of the principal architects of the science of statistical quality control said, “You cannot inspect quality into a product.” In other words, once the inspection takes place, it’s too late. Rather, data from the quality inspection needs to be utilized to continually improve the process.

Businessman Philip B. Crosby, who developed the concept of Zero Defects while working as senior quality engineer at aircraft manufacturer The Martin Company, defined quality as “a conformance to requirements.” He warned against the high cost of nonconformance and said that the desired performance standard of zero defects could only be achieved through the proper management system.

Management consultant Joseph Juran, who focused on management training and the human element of quality control for a variety of businesses, stated that quality is “a fitness for use.”

¹⁵⁰ Wikipedia, *Lean Six Sigma*, https://en.wikipedia.org/wiki/Lean_Six_Sigma

¹⁵¹ Wikipedia, *Six Sigma*, https://en.wikipedia.org/wiki/Six_Sigma

¹⁵² Wikipedia, *Lean Manufacturing*, https://en.wikipedia.org/wiki/Lean_manufacturing

¹⁵³ YouTube.com, *A Theory of a System for Educators and Managers*, <https://www.youtube.com/watch?v=2MJ3IGJ4OFo>

Understanding Systems

The following are several definitions of the term system:

- A system is a set of connected things or parts forming a complex whole.
- A system is a regularly interacting or interdependent group of items forming a unified whole.
- A system is a set of detailed methods, procedures and routines created to carry out a specific activity, perform a duty, achieve an objective, or solve a problem.

Deming explains that, “the typical way of managing a complex system is to take the system, break it into parts, and then try and manage each part as well as possible.” But that does not work because it is possible to improve the performance of each part, and destroy the system as a whole.

Per Deming, “Working together is the main contribution to systemic thinking as opposed to working apart separately.”

You cannot explain the behavior of a system by analyzing the system; you have to use synthesis to understand a system. There is a difference between analysis and synthesis:

- **Analysis** is separate the whole into parts and study each part individually. Analysis is the dominate mode of thought in the western world. You cannot explain the behavior of a system by analysis. You can reveal its structure and see how it works, but you cannot understand why it works the way it works.
- **Synthesis** the combination of ideas to form a theory or system. If you want to understand why something works the way it does you use synthesis to figure that out.

If you want to understand how something works you use analysis. If you want to understand why something works the way it does you use synthesis. Both analysis and synthesis is necessary to understand a system.

So, a **system**¹⁵⁴ is a regularly interacting or interdependent group of items forming a unified whole. A **system of systems**¹⁵⁵ is a system where the elements and sub-assemblies of a system are themselves systems. A system of systems is a complex system or a set of interconnected and coordinated autonomous systems to satisfy specific capacity and functions that the systems could not achieve independently.

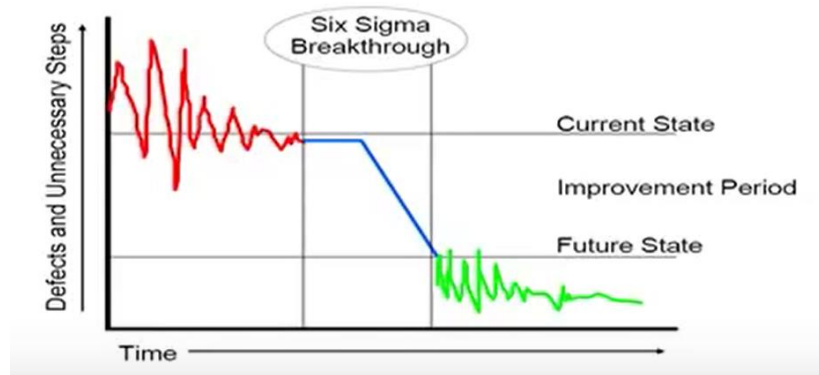
¹⁵⁴ Wikipedia, *System*, <https://en.wikipedia.org/wiki/System>

¹⁵⁵ Wikipedia, *System of Systems*, https://en.wikipedia.org/wiki/System_of_systems

Lean Six Sigma Principles

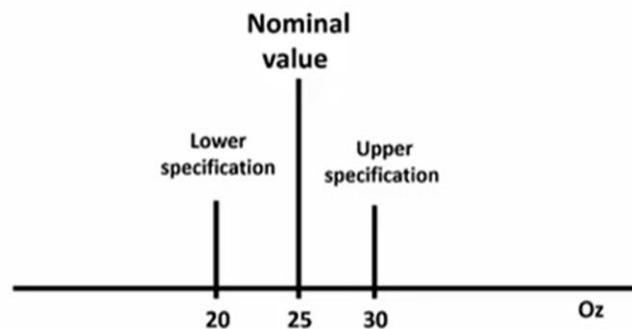
Six sigma¹⁵⁶ is about reducing variability and increasing predictability. These are the fundamental principles of Lean Six Sigma:

- Reducing variability
- Reducing defects
- Reducing unnecessary steps
- Improving predictability



Process Capability

Process capability¹⁵⁷ is the ability of a process to repeatedly meet a specified requirement.



Methodology and Process for Moving Information

Moving information is a process. What are the capabilities of your accounting information system? What alternative processes are available and what are their capabilities?

An accounting information system is a system that collects, records, stores, and processes data to produce information for decision makers within an economic entity. Accounting information systems generally consist of six primary components:

¹⁵⁶ YouTube.com, *What Is Six Sigma? The basic principles of Six Sigma*, <https://www.youtube.com/watch?v=C8g-Zzgk1Uw>

¹⁵⁷ YouTube.com, *Process capability and process capability index*, <https://www.youtube.com/watch?v=ouMnoLcK6uw>

- **People** who use the system,
- **Procedures** and instructions used,
- **Data** (or information) about the economic entity and its activities and transactions,
- **Software** used to process the data/information,
- **Hardware**, or information technology infrastructure, used by the system,
- **Internal controls and security measures** used to safeguard the information and assets of the economic entity.

The business function of an accounting information system is to collect and store data about organizational activities, resources, and personnel; transform data into information so management can plan, execute, control, evaluate activities, resources, and personnel; and to provide adequate controls to safeguard the organization's assets and information.

Problems with Spreadsheets

Many companies use spreadsheets within their accounting information systems extensively. But there are problems with spreadsheets¹⁵⁸. Here is a summary of the top 10 disadvantages of spreadsheets¹⁵⁹:

1. Vulnerable to fraud
2. Susceptible to trivial human errors
3. Difficult to troubleshoot or test
4. Obstructive to regulatory compliance
5. Unfit for agile business practices
6. Not designed for collaborative work
7. Hard to consolidate
8. Incapable of supporting quick decision making
9. Unsuitable for business continuity
10. Scales poorly

Poka-yoke (Mistake proofing)

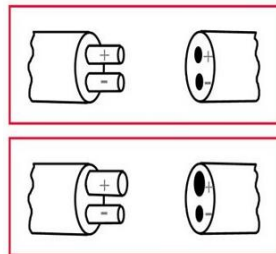
Poka-yoke is a technique used to prevent mistakes through smarter design. Poka-yoke¹⁶⁰ is a Japanese term that means "mistake-proofing". A poka-yoke is any mechanism consciously added to a process that helps an equipment operator avoid mistakes. Its purpose is to eliminate defects by preventing, correcting, or drawing attention to human errors as the errors occur.

¹⁵⁸ YouTube.com, Problem with Spreadsheets, <https://www.youtube.com/watch?v=wbiVK6HKHHg>

¹⁵⁹ Denizon Team, *Top 10 Disadvantages of Spreadsheets*, <http://www.denizon.com/spreadsheets/top-10-disadvantages-of-spreadsheets/>

¹⁶⁰ Wikipedia, *Poka-yoke*, <https://en.wikipedia.org/wiki/Poka-yoke>

For example, consider the graphic¹⁶¹ below. You want someone to insert a plug into the receptacle such that positive and negative match up; inadvertently reversing this would have catastrophic consequences. In the top graphic, notice that it is possible to make a mistake but in the bottom a mistake would be impossible because of the size differences in the positive and negative receptacle and plug.



Smart design means less user errors. Blocks are a mechanism for implementing poka-yoke, or mistake proofing XBRL-based information. Primitive object structure, mechanical relations, mathematical relations, logical relations, and even some accounting relations must make sense relative to other primitive objects. Blocks and the structured nature of XBRL make implementing these mistake proofing techniques possible with financial report creation software.

Double-entry accounting is a type of poka-yoke mechanism used by professional accountants. The first recorded use of double-entry accounting was in 1211 AD by a bank in Florence¹⁶². The foundational basis of double-entry accounting is straightforward. Quoting David Ellerman from his paper *The Math of Double-Entry Bookkeeping: Part I (scalars)*¹⁶³:

“Given an equation $w + \dots + x = y + \dots + z$, it is not possible to change just one term in the equation and have it still hold. Two or more terms must be changed.”

And so, the left hand side of the equation “ $w + \dots + x$ ” (the DEBIT side) must always equal the right hand side of the equation “ $y + \dots + z$ ” (the CREDIT side) in double-entry accounting. The reason that double-entry accounting is used, as contrast to single-entry accounting, is double-entry accounting’s capability to detect errors and to distinguish an error from fraud. Double-entry accounting is smart design.

¹⁶¹ Process Exam, *Six Sigma Tools - Poka Yoke*, <http://www.processexam.com/six-sigma-tools-poka-yoke>

¹⁶² Geoffrky Alan Lee, *The Development of Italian Bookkeeping 1211–1300*, Wiley, <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1467-6281.1973.tb00183.x>

¹⁶³ David Ellerman, *The Math of Double-Entry Bookkeeping: Part I (scalars)*, <http://www.ellerman.org/the-math-of-double-entry-bookkeeping-part-i-scalars/>

Excessive Process Automation

Don't overdo it with process automation. First, only processes that are working effectively should be automated. Automating a process that is not working makes no sense. Second, take an incremental approach to automating processes.

Tesla's CEO Elon Musk said, "Excessive automation at Tesla was a mistake." In the 1980s, General Motors wasted billions of dollars in a largely fruitless effort to automate car product. A lot of mistakes like these are made only to be repeated in the future because people are not aware of the history.

Automation works best when it is applied incrementally to a process that is already working smoothly and effectively.

Conclusion

Professional accountants need to understand how to think critically, systematically, deliberately, rigorously, and methodically about how computers work and how to get computers to do what you want them to do. Machine intelligence is real and a tool that can, and will, be leveraged by professional accountants. Human and machine collaboration is how work will be performed in the future.

Professional accountants already have many of the skills that they need such as procedural thinking, systematic problem solving, and systems thinking. They need to add to those skills a few other skills such as technological empathy or understanding how a computer sees the world and abstract thinking.

Next steps

The next step is to dig into the next level of detail. That next level can be found in ***Intelligent XBRL-based Digital Financial Reporting***¹⁶⁴.

¹⁶⁴ Charles Hoffman and Rene van Egmond, *Intelligent XBRL-based Digital Financial Reporting*, <http://xbml.squarespace.com/intelligent-xml/>